



**SAVEETHA SCHOOL OF ENGINEERING,  
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL  
SCIENCES**



**TRANSMISSION LINES  
&  
WAVEGUIDES**

**ECA15**

**NAME:**

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**SAVEETHA SCHOOL OF ENGINEERING**  
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|                |                                                                                                                                                                |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EXNO: 1</b> | <b>Estimation of Characteristic Impedance (<math>Z_0</math>), and Propagation Constant (<math>\gamma</math>) of a Transmission Line from Primary Constants</b> |
| <b>DATE:</b>   |                                                                                                                                                                |

**Test Case 1:**

**Investigating the significance of Characteristic Impedance ( $Z_0$ ), and Propagation Constant ( $\gamma$ ) of a Transmission Line using the following parameters**

- a) Resistance =  $10 \Omega/\text{Km}$
- b) Capacitance =  $8 \mu\text{F}/\text{Km}$
- c) Inductance =  $3.5 \text{ mH}/\text{Km}$
- d) Conductance =  $4 \mu\text{mho}/\text{Km}$
- e) Frequency =  $2 \text{ Khz}$

**AIM:** Investigating the significance of Characteristic Impedance ( $Z_0$ ), and Propagation Constant ( $\gamma$ ) of a Transmission Line with Resistance =  $10 \Omega/\text{Km}$ , Capacitance =  $8 \mu\text{F}/\text{Km}$ , Inductance =  $3.5 \text{ mH}/\text{Km}$ , Conductance =  $4 \mu\text{mho}/\text{Km}$ , Frequency =  $2 \text{ Khz}$

**SOFTWARE REQUIRED:**

SCILAB Version 6.1.1

**PROCEDURE:**

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

**CALCULATION:**

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{\frac{Z}{Y}}$$

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)} = \sqrt{ZY}$$

### PROGRAM:

```
clear; clc;
R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
w=2*pi*f;
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);
```

### OUTPUT:

Constants L=3mH; f=2000; G=.4\*(10^-6); C=8\*(10^-6).

| Simulation Results |                     |                      | Theoretical Results |                      |
|--------------------|---------------------|----------------------|---------------------|----------------------|
| Z0(ohms)           | $\alpha$ (neper/km) | $\beta$ (radians/km) | $\alpha$ (neper/km) | $\beta$ (radians/km) |
|                    |                     |                      |                     |                      |

### RESULT:

The characteristic impedance and propagation constant of a transmission line is calculated using SCILAB and the impact of primary constants on secondary constants are investigated.

### Test Case 2:

Examine the impact of resistance and inductance on the significance of Characteristic Impedance ( $Z_0$ ) and Propagation Constant ( $\gamma$ ) of a Transmission Line at a frequency  $f=2.5$  Khz. Vary the resistance value from  $10\ \Omega$  to  $50\ \Omega$  at the interval of  $10\ \Omega$  with L,G and C values constant and calculate the Characteristic Impedance and Propagation Constant . Vary the inductance values from 3.5 mH to 11.5 mH at the interval of 2.5 mH with R,G and C values constant.

**AIM:** To examine the impact of resistance and inductance on the significance of Characteristic Impedance ( $Z_0$ ) and Propagation Constant ( $\gamma$ ) of a Transmission Line at a frequency  $f=2.5$  KHz. Vary the resistance value from  $10\ \Omega$  to  $50\ \Omega$  at the interval of  $10\ \Omega$  with L,G and C values constant and calculate the Characteristic Impedance and Propagation Constant . Vary the inductance values from  $3.5\text{ mH}$  to  $11.5\text{ mH}$  at the interval of  $2.5\text{ mH}$  with R,G and C values constant.

### SOFTWARE REQUIRED:

SCILAB Version 6.1.1

### PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program
- 

### CALCULATION:

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{\frac{Z}{Y}}$$

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)} = \sqrt{ZY}$$

### PROGRAM:

```
clear;clc;
R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
w=2*pi*f;
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
```

```
printf('-Phase constant b = %f radians/km',b1);
```

## OUTPUT:

Variable R

Constants  $L=3\text{mH}$ ;  $f=2000$ ;  $G=.4*(10^{-6})$ ;  $C=8*(10^{-6})$ .

| S. No. | Inputs (R) | Simulation Results |                           |                            | Theoretical Results       |                            |
|--------|------------|--------------------|---------------------------|----------------------------|---------------------------|----------------------------|
|        |            | $Z_0(\text{ohms})$ | $\alpha(\text{neper/km})$ | $\beta(\text{radians/km})$ | $\alpha(\text{neper/km})$ | $\beta(\text{radians/km})$ |
|        |            |                    |                           |                            |                           |                            |
|        |            |                    |                           |                            |                           |                            |
|        |            |                    |                           |                            |                           |                            |
|        |            |                    |                           |                            |                           |                            |
|        |            |                    |                           |                            |                           |                            |

Variable L

Constants  $R=10\Omega$ ;  $f=2000$ ;  $G=.4*(10^{-6})$ ;  $C=8*(10^{-6})$ .

| S. No. | Inputs (L) | Simulation Results |                           |                            | Theoretical Results       |                            |
|--------|------------|--------------------|---------------------------|----------------------------|---------------------------|----------------------------|
|        |            | $Z_0(\text{ohms})$ | $\alpha(\text{neper/km})$ | $\beta(\text{radians/km})$ | $\alpha(\text{neper/km})$ | $\beta(\text{radians/km})$ |
|        |            |                    |                           |                            |                           |                            |
|        |            |                    |                           |                            |                           |                            |
|        |            |                    |                           |                            |                           |                            |
|        |            |                    |                           |                            |                           |                            |
|        |            |                    |                           |                            |                           |                            |

## RESULT:

The impact of resistance and inductance on the significance of Characteristic Impedance ( $Z_0$ ) and Propagation Constant ( $\gamma$ ) of a Transmission Line is examined at a frequency  $f=2.5\text{ KHz}$ .

## Test Case 3:

**Explore the impact of frequency and capacitance on the significance of Characteristic Impedance ( $Z_0$ ) and Propagation Constant ( $\gamma$ ) of a Transmission Line. Vary the frequency from 2KHz to 10 KHz at the interval of 2 KHz with L,R ,G and C values constant and calculate the Characteristic Impedance and Propagation Constant . Vary the capacitance values from 8  $\mu\text{F}$  to 16  $\mu\text{F}$  at the interval of 2  $\mu\text{F}$  with R,G and L values constant.**

**AIM:** To explore the impact of frequency and capacitance on the significance of Characteristic Impedance ( $Z_0$ ) and Propagation Constant ( $\gamma$ ) of a Transmission Line. Vary the frequency from 2KHz to 10 KHz at the interval of 2 KHz with L,R ,G and C values constant and calculate the Characteristic Impedance and Propagation Constant

### CALCULATION:

$$Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}} = \sqrt{\frac{Z}{Y}}$$

$$\gamma = \sqrt{(R+j\omega L)(G+j\omega C)} = \sqrt{ZY}$$

### PROGRAM:

```
clear; clc;
R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
w=2*pi*f;
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);
```

### OUTPUT:

Variable f

Constants L=3mH; f=2000; G=.4\*(10^-6); C=8\*(10^-6).

| S. No. | Inputs<br>(f) Hz | Simulation Results |                        |                         | Theoretical Results    |                         |
|--------|------------------|--------------------|------------------------|-------------------------|------------------------|-------------------------|
|        |                  | Z0(ohms)           | $\alpha$<br>(neper/km) | $\beta$<br>(radians/km) | $\alpha$<br>(neper/km) | $\beta$<br>(radians/km) |
|        |                  |                    |                        |                         |                        |                         |
|        |                  |                    |                        |                         |                        |                         |
|        |                  |                    |                        |                         |                        |                         |
|        |                  |                    |                        |                         |                        |                         |
|        |                  |                    |                        |                         |                        |                         |

Variable C

Constants  $R=10$ ;  $L=.009$ ;  $f=2500$ ;  $G=.4 \times 10^{-6}$ ;

| S.<br>No. | Inputs<br>(C) F | Simulation Results |                            |                             | Theoretical Results    |                             |
|-----------|-----------------|--------------------|----------------------------|-----------------------------|------------------------|-----------------------------|
|           |                 | $Z_0(\text{ohms})$ | $\alpha$<br>(neper/k<br>m) | $\beta$<br>(radians/k<br>m) | $\alpha$<br>(neper/km) | $\beta$<br>(radians/k<br>m) |
|           |                 |                    |                            |                             |                        |                             |
|           |                 |                    |                            |                             |                        |                             |
|           |                 |                    |                            |                             |                        |                             |
|           |                 |                    |                            |                             |                        |                             |
|           |                 |                    |                            |                             |                        |                             |

### RESULT:

The impact of frequency and capacitance on the significance of Characteristic Impedance ( $Z_0$ ) and Propagation Constant ( $\gamma$ ) of a Transmission Line is investigated for various values of frequencies and capacitance values.

|                |                                                                                                                              |
|----------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>EXNO: 2</b> | <b>Analysis of SWR (Standing Wave Ratio) Transmission Lines by varying<br/>a) load impedance b) characteristic impedance</b> |
| <b>DATE:</b>   |                                                                                                                              |

#### Test Case 1:

Analyze the SWR of a transmission line for a fixed characteristic impedance ( $Z_0$ ) of 50 ohms, as the load impedance ( $Z_L$ ) is varied across 650 ohms to 850 ohms. Verify how the SWR changes with each value of  $Z_L$ . When  $Z_L = Z_0$ , the SWR should equal 1 (indicating no reflection).

**AIM:** To analyze the SWR of a transmission line for a fixed characteristic impedance ( $Z_0$ ) of 50 ohms, as the load impedance ( $Z_L$ ) is varied across the following values: 25 ohms, 50 ohms, 75 ohms, and 100 ohms. To verify how the SWR changes with each value of  $Z_L$  and when  $Z_L = Z_0$ .

#### SOFTWARE REQUIRED:

SCILAB Version 6.1.1

#### PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

#### CALCULATION

$$K = \frac{Z_R - Z_0}{Z_R + Z_0}$$

$$S = \frac{1 + |K|}{1 - |K|}$$

#### PROGRAM:

```
clc;clear;close;
Zo=200;ZR=650-(%i*475);
K=(ZR-Zo)/(ZR+Zo);
```

```
ampK=sqrt((real(K)^2)+(imag(K)^2));S=(1+ampK)/(1-ampK);
printf("(c)Standing wave ratio = %f",round(S*1000)/1000);
```

### OUTPUT:

**Zo=50 ohms**Variable:ZR

| ZROhms | Simulation Results | Theoretical Results |
|--------|--------------------|---------------------|
|        | SWR                | SWR                 |
|        |                    |                     |
|        |                    |                     |
|        |                    |                     |
|        |                    |                     |
|        |                    |                     |
|        |                    |                     |

**ZR=50/\_40oohms**Variable:Zo

| Zo Ohms | Simulation Results | Theoretical Results |
|---------|--------------------|---------------------|
|         | SWR                | SWR                 |
|         |                    |                     |
|         |                    |                     |
|         |                    |                     |
|         |                    |                     |
|         |                    |                     |
|         |                    |                     |

### RESULT:

The SWR of a transmission line for a fixed characteristic impedance (Zo) of 50 ohms, is analyzed for various values of load impedance ZL. SWR is also examined for the condition  $Z_L = Z_o$ .

### Test Case 2:

**Analyze the SWR for a fixed load impedance ( $Z_L$ ) of 50 ohms while varying the characteristic impedance ( $Z_0$ ) of the transmission line to 250 ohms to 400 ohms. Verify how the SWR changes with each value  $Z_0$  including  $Z_0=Z_L$ .**

**AIM:** To analyze the SWR for a fixed load impedance ( $Z_L$ ) of 75 ohms while varying the characteristic impedance ( $Z_0$ ) of the transmission line to 25 ohms, 50 ohms, 75 ohms, and 100 ohms. To Verify how the SWR changes with each value  $Z_0$  including  $Z_0=Z_L$

### **SOFTWARE REQUIRED:**

SCILAB Version 6.1.1

### **PROCEDURE:**

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program
- End the program

### **CALCULATION**

$$K = \frac{Z_R - Z_0}{Z_R + Z_0}$$

$$S = \frac{1 + |K|}{1 - |K|}$$

### **PROGRAM:**

```
clc;clear;close;  
Zo=200;ZR=650-(%i*475);  
K=(ZR-Zo)/(ZR+Zo);  
ampK=sqrt((real(K)^2)+(imag(K)^2));S=(1+ampK)/(1-ampK);  
printf("(c)Standing wave ratio = %f",round(S*1000)/1000);
```

### **OUTPUT:**

ZR=50/\_40° ohms Variable: Zo

| Zo Ohms | Simulation Results | Theoretical Results |
|---------|--------------------|---------------------|
|         | SWR                | SWR                 |
|         |                    |                     |
|         |                    |                     |
|         |                    |                     |
|         |                    |                     |
|         |                    |                     |

### RESULT:

SWR for a fixed load impedance ( $Z_L$ ) of 50 ohms is analyzed while varying the characteristic impedance ( $Z_0$ ) of the transmission line to 250 ohms to 400 ohms. Impact of the condition  $Z_0=Z_L$  on SWR is also examined for various values of  $Z_0$ .

|                |                                                                                                                                                                       |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EXNO: 3</b> | <b>Designing a Quarter wave transformer by varying following parameters</b><br><b>a) frequency</b><br><b>b) input impedance</b><br><b>c) characteristic impedance</b> |
| <b>DATE:</b>   |                                                                                                                                                                       |

**Designing a Quarter wave transformer by varying following parameters**

**d) frequency**

**e) input impedance**

**f) characteristic impedance**

**Experiment Number:7**

**Test Case 1:**

**Design a quarter-wave transformer for an input impedance ( $Z_{in}$ ) of 50 ohms and a load impedance ( $Z_L$ ) of 100 ohms. Analyze the transformer's performance as the operating frequency is varied across {100 MHz, 200 MHz, 300 MHz, 500 MHz}.**

AIM: To design a quarter-wave transformer for an input impedance ( $Z_{in}$ ) of 50 ohms and a load impedance ( $Z_L$ ) of 100 ohms and to analyze the transformer's performance as the operating frequency is varied across {100 MHz, 200 MHz, 300 MHz, 500 MHz }.

**SOFTWARE REQUIRED:**

SCILAB

**PROCEDURE:**

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

**CALCULATION:**

$$R_0 = \sqrt{Z_R \cdot Z_{in}}$$

# **PROGRAM:**

```
clear;clc;
Zr=100;s=9;d=0.1;
Zs=50;
Ro=sqrt(Zs*Zr);f=500*(10^6);r=d/2;
Zof=276*log10(s/r);
Zoq=sqrt(Zr*Zof);
L=300/(f*(10^-6));
printf("length of lambda/4 transformer = %f meters\n",L);
printf("characteristic impedance of quarter wave line = %f ",fix(Ro*10000)/10000);
```

# **OUTPUT:**

**Z<sub>R</sub>=100 ohms Z<sub>in</sub> =50 ohms**

| <b>Z<sub>S</sub> ohms</b> | <b>Z<sub>R</sub> ohms</b> | <b>R<sub>o</sub> ohms</b> |
|---------------------------|---------------------------|---------------------------|
|                           |                           |                           |

**Z<sub>R</sub>=100 ohms Z<sub>in</sub> =50 ohms**

| <b>Frequency (MHz)</b> | <b>Length of λ/4 transformer (meters)</b> |
|------------------------|-------------------------------------------|
|                        |                                           |
|                        |                                           |
|                        |                                           |
|                        |                                           |

**RESULT:** A quarter-wave transformer for an input impedance (Z<sub>in</sub>) of 50 ohms and a load impedance (Z<sub>L</sub>) of 100 ohms is designed and the transformer's performance is

analyzed for the varying operating frequency is varied across {100 MHz, 200 MHz, 300 MHz, 500 MHz }.

### **Test Case 2:**

**For a fixed operating frequency of 2 GHz and a load impedance ( $Z_L$ ) of 100 ohms, design a quarter-wave transformer and evaluate its performance as the input impedance ( $Z_{in}$ ) varies through {25 ohms, 50 ohms, 75 ohms, 100 ohms}. Validate that the transformer's characteristic impedance ( $Z_{\text{transformer}} = R_0 = \sqrt{Z_{in} \cdot Z_L}$ ) changes with  $Z_{in}$ .**

AIM: To design a quarter-wave transformer a fixed operating frequency of 2 GHz and a load impedance ( $Z_L$ ) of 100 ohms and evaluate its performance as the input impedance ( $Z_{in}$ ) varied .To validate that the transformer's characteristic impedance ( $Z_{\text{transformer}} = R_0 = \sqrt{Z_{in} \cdot Z_L}$ ) changes with  $Z_{in}$ .

### **SOFTWARE REQUIRED:**

SCILAB

### **PROCEDURE:**

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

### **CALCULATION**

$$R_0 = \sqrt{Z_R \cdot Z_{in}}$$

### **PROGRAM:**

```
clear;clc;
Zr=100;s=9;d=0.1;
Zs=50;
Ro=sqrt(Zs*Zr);f=100*(10^6);r=d/2;
Zof=276*log10(s/r);
Zoq=sqrt(Zr*Zof);
```

```

L=300/(f*(10^-6));
printf("length of lambda/4 transformer = %f lambda\n",L);
printf("characteristic impedance of quarter wave line = %f ",fix(Ro*10000)/10000);

```

## OUTPUT:

**ZR=100 ohms Variable :Zin**

| <b>Z<sub>in</sub></b> | <b>Ro</b> |
|-----------------------|-----------|
|                       |           |
|                       |           |
|                       |           |
|                       |           |

**RESULT:** A quarter-wave transformer is designed at a fixed operating frequency of 2 GHz and load impedance ( $Z_L$ ) of 100 ohms and its performance is evaluated for various input impedance values. The variation of transformer's characteristic impedance with  $Z_{in}$  is also validated..

|                |                                                                                                                                                                                    |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EXNO: 4</b> | <b>Strategize impedance matching with characteristic impedance and load impedance through the measurement of</b><br><b>a) Location of the stub</b><br><b>b) Length of the stub</b> |
| <b>DATE:</b>   |                                                                                                                                                                                    |

### Test Case 1:

Optimize impedance matching using single stub matching by finding the location and length of the stub for the values of load impedance of  $100 \angle 30^\circ$  to  $350 \angle 30^\circ$  keeping the characteristic impedance  $Z_0 = 60 \Omega$  constant.

AIM: To optimize impedance matching using single stub matching by finding the location and length of the stub for the values of load impedance of  $100 \angle 30^\circ$  to  $350 \angle 30^\circ$  keeping the characteristic impedance  $Z_0 = 60 \Omega$  constant.

### SOFTWARE REQUIRED:

SCILAB

### PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

### CALCULATION

$$s_1 = \frac{\phi + \pi - \cos^{-1}(|K|)}{2\beta}$$

Length of the stub,

$$L = \frac{\lambda}{2\pi} \tan^{-1} \left[ \frac{\sqrt{1-|K|^2}}{2|K|} \right]$$

## PROGRAM:

```
clear; clear; clc;
ZR=100;Zo=600;f=100*(10^6);
lo=300/(f*(10^-6)); //lo=wavelength Ls=(lo/(2*pi))*(atan(sqrt(ZR/Zo)));
printf("-Point of attachment = %f cms\n",round(Ls*(10^2)*10)/10)
Lt=(lo/(2*pi))*(pi+atan((sqrt(ZR*Zo))/(ZR-Zo)));
printf("-Length of the short circuited stub = %f cms",round(Lt*(10^2)));
```

## OUTPUT:

Variables  $Z_R$ ;  
Constants  $Z_o = 60\Omega$ ;  $f=100*(10^6)$ ;

| Input                 | Output              |                                        |
|-----------------------|---------------------|----------------------------------------|
| ZR<br>Ohms. angle 30° | Point of attachment | Length of short-circuited<br>stub (cm) |
|                       |                     |                                        |
|                       |                     |                                        |
|                       |                     |                                        |
|                       |                     |                                        |
|                       |                     |                                        |

## RESULT:

f Impedance matching using single stub matching is optimized by calculating the location and length of the stub or varying values of load impedance keeping the characteristic impedance  $Z_o = 60\Omega$  constant .

## Test Case 2:

Optimize impedance matching using single stub matching by finding the location and length of the stub for the values of characteristic impedance ranging from  $600\Omega$  to  $1800\Omega$ . Keeping the load impedance  $Z_R = 100 \angle 30^\circ$  constant .

AIM: To optimize impedance matching using single stub matching by finding the location and length of the stub for the values of characteristic impedance ranging from  $600\Omega$  to  $1800\Omega$ . Keeping the load impedance  $Z_R = 100 \angle 30^\circ$  constant .

## SOFTWARE REQUIRED:

SCILAB

## PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

## CALCULATION

$$s_1 = \frac{\phi + \pi - \cos^{-1}(|K|)}{2\beta}$$

Length of the stub,

$$L = \frac{\lambda}{2\pi} \tan^{-1} \left[ \frac{\sqrt{1-|K|^2}}{2|K|} \right]$$

## PROGRAM:

```
clear; clear; clc;
```

```
ZR=100;Zo=600;f=100*(10^6);
```

```
lo=300/(f*(10^-6)); //lo=wavelength Ls=(lo/(2*pi))*(atan(sqrt(ZR/Zo)));
```

```
printf("-Point of attachment = %f cms\n",round(Ls*(10^2)*10)/10)
```

```
Lt=(lo/(2*pi))*(pi+(atan((sqrt(ZR*Zo))/(ZR-Zo))));
```

```
printf("-Length of the short circuited stub = %f cms",round(Lt*(10^2)));
```

## OUTPUT:

Variables  $Z_0$ ;

Constants  $Z_R = 100 \angle 30^\circ \Omega$ ;  $f = 100 \times (10^6)$ ;

| Input          | Output              |                                     |
|----------------|---------------------|-------------------------------------|
| $Z_0$<br>Ohms. | Point of attachment | Length of short-circuited stub (cm) |
|                |                     |                                     |
|                |                     |                                     |
|                |                     |                                     |
|                |                     |                                     |
|                |                     |                                     |

## RESULT:

Impedance matching using single stub matching is optimized by calculating the location and length of the stub for varying values of characteristic impedance keeping the load impedance constant .

|                |                                                                                                               |
|----------------|---------------------------------------------------------------------------------------------------------------|
| <b>EXNO: 5</b> | <b>Calculation of Group velocity, Phase velocity and phase constant of EM waves in rectangular waveguides</b> |
| <b>DATE:</b>   |                                                                                                               |

**Test Case 1 :**

You are experimenting with a rectangular waveguide to analyze the relationship between its dimensions and wave propagation. The waveguide dimensions are a=4.0 cm, b=2.0 cm and the operating frequency is f=8 GHz.

1. Calculate the cutoff frequencies for TE<sub>10</sub> modes.
2. Predict the phase and group velocities for the TE<sub>10</sub> mode at the given frequency.
3. If the width of the waveguide is reduced to a=3.0 cm, how does it affect the cutoff frequency and propagation characteristics?

**AIM:** To find the group velocity, phase velocity and phase constant of Electromagnetic waves in Rectangular Waveguides whose dimensions are a=4.0 cm, b=2.0 cm and the operating frequency is f=8 GHz.

**SOFTWARE REQUIRED:**

SCILAB Version 6.1.1

**PROCEDURE:**

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

**CALCULATION**

$$\beta = \sqrt{\omega^2 \mu \epsilon - \left(\frac{\pi}{a}\right)^2} = \omega \sqrt{\mu \epsilon} \sqrt{1 - \left(\frac{f_c}{f}\right)^2}.$$

$$\text{Group velocity } v_g = c \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

$$\text{Phase velocity } v_p = \frac{c}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$$

#### PROGRAM:

```
clc;
clear;
close;
//TE10 Mode
a=8.636*10^-2;
b=4.318*10^-2;
f=4*10^9;
V=3*10^8;
m=input("enter the value of m= "); n=input("enter the value of n= ");
fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);
disp("Cut off frequency=",fc*10^-9);
//let phase velocity = A
A=V/sqrt(1-(fc/f)^2);
disp("Phase velocity in Mm/sec=",A*10^-6);
//let group velocity = B
B=V*A/A;
disp("Group velocity in Mm/sec =", B*10^-6);
```

#### OUTPUT:

##### Case 1:

**Original Dimensions (a=4.0 cm,b=2.0 cm)**

##### Inputs:

- m=1, n=0.
- f=8 GHz.

**Outputs:**

| Parameter             | Simulation Results | Theoretical Results |
|-----------------------|--------------------|---------------------|
|                       | Value              | Value               |
| Cutoff Frequency (fc) |                    |                     |
| Phase Velocity (vp)   |                    |                     |
| Group Velocity (vg)   |                    |                     |

**Case 2:**

Reduced Width (a=3.0 cm,b=2.0 cm)

**Inputs:**

- m=1, n=0.
- f=8 GHz.

**Outputs:**

| Parameter             | Simulation Results | Theoretical Results |
|-----------------------|--------------------|---------------------|
|                       | Value              | Value               |
| Cutoff Frequency (fc) |                    |                     |
| Phase Velocity (vp)   |                    |                     |
| Group Velocity (vg)   |                    |                     |

**Table: Variation of Phase and Group Velocities with a**

| Waveguide Width (a) | Simulation Results    |                     |                     | Theoretical Results   |                     |                     |
|---------------------|-----------------------|---------------------|---------------------|-----------------------|---------------------|---------------------|
|                     | Cutoff Frequency (fc) | Phase Velocity (vp) | Group Velocity (vg) | Cutoff Frequency (fc) | Phase Velocity (vp) | Group Velocity (vg) |
| 4.0 cm              |                       |                     |                     |                       |                     |                     |
| 3.5 cm              |                       |                     |                     |                       |                     |                     |
| 3.0 cm              |                       |                     |                     |                       |                     |                     |
| 2.5 cm              |                       |                     |                     |                       |                     |                     |
| 2.0 cm              |                       |                     |                     |                       |                     |                     |

**RESULT:**

The group velocity, phase velocity and phase constant of waves in a rectangular waveguide are estimated

using SCILAB.

### Test Case 2 :

In a radar system operating at  $f=15$  GHz the waveguide dimensions are  $a = 2.5$  and  $b = 1.25$  cm

1. Compute the cutoff frequencies for  $TE_{10}$  modes.
2. Determine the phase and group velocities for the  $TE_{10}$  mode.
3. If dispersion is critical to radar signal accuracy, what strategies would you recommend maintaining signal fidelity?

**AIM:** To find the group velocity, phase velocity and phase constant of Electromagnetic waves in Rectangular Waveguides whose dimensions are  $a= 2.5$  cm,  $b=1.25$  cm and the operating frequency is  $f=15$  GHz.

### SOFTWARE REQUIRED:

SCILAB Version 6.1.1

### PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

### CALCULATION

### PROGRAM:

```
clc;
clear;
close;
//TE10 Mode
a=8.636*10^-2;
b=4.318*10^-2;
f=4*10^9;
V=3*10^8;
m=input("enter the value of m= "); n=input("enter the value of n= ");
fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);
disp("Cut off frequency=",fc*10^-9);
//let phase velocity = A
```

```

A=V/sqrt(1-(fc/f)^2);
disp("Phase velocity in Mm/sec=",A*10^-6);
//let group velocity = B
B=V*V/A;
disp("Group velocity in Mm/sec =", B*10^-6);

```

### Program Output for TE<sub>10</sub> Mode:

Using the program with a=0.025 m b=0.0125 m and f=15 GHz:

| Parameter                  | Simulation Results | Theoretical Results |
|----------------------------|--------------------|---------------------|
|                            | Value              | Value               |
| Cutoff Frequency ( $f_c$ ) |                    |                     |
| Phase Velocity ( $v_p$ )   |                    |                     |
| Group Velocity ( $v_g$ )   |                    |                     |

### RESULT:

The group velocity, phase velocity and phase constant of waves in a rectangular waveguide are estimated using SCILAB.

### Test Case 3 :

A waveguide with dimensions a=5.0 cm and b=2.5 cm is used to carry a signal at f=10 GHz

1. Determine the number of TE modes that can propagate at this frequency.
2. For the TE<sub>10</sub> mode, calculate the phase velocity and group velocity.
3. Discuss how the presence of multiple propagating modes affects the transmission quality and suggest practical solutions to limit multimode effects.

### AIM:

To find the group velocity, phase velocity and phase constant of Electromagnetic waves in Rectangular Waveguides whose dimensions are a= 5.0 cm, b=2.5 cm and the operating frequency is f=10 GHz.

### SOFTWARE REQUIRED:

SCILAB Version 6.1.1

**PROCEDURE:**

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

**CALCULATION****PROGRAM:**

```
clc;
clear;
close;
//TE10 Mode
a=8.636*10^-2;
b=4.318*10^-2;
f=4*10^9;
V=3*10^8;
m=input("enter the value of m= ");
n=input("enter the value of n= ");
fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);
disp("Cut off frequency=",fc*10^-9);
//let phase velocity = A
A=V/sqrt(1-(fc/f)^2);
disp("Phase velocity in Mm/sec=",A*10^-6);
//let group velocity = B
B=V*V/A;
disp("Group velocity in Mm/sec =", B*10^-6);
```

**OUTPUT:****Waveguide Parameters:**

- Width (a) = 5.0 cm=0.05m

- Height (b) = 2.5 cm=0.025 m
- Operating Frequency (f) = 10 GHz
- Speed of Light (V) =  $3 \times 10^8$  m/s

### 1. Number of TE Modes that Can Propagate:

The cutoff frequency for a  $TE_{mn}$  mode is:

$$f_c(m,n) = V/2 \left( \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \right)$$

Modes propagate if  $f > f_c(m,n)$ . The number of modes can be determined by calculating  $f_c$  for different combinations of m and n.

|                  | Simulation Results    |                      | Theoretical Results   |                      |
|------------------|-----------------------|----------------------|-----------------------|----------------------|
| Mode (m,n)       | Cutoff Frequency (fc) | Propagation (Yes/No) | Cutoff Frequency (fc) | Propagation (Yes/No) |
| TE <sub>10</sub> |                       |                      |                       |                      |
| TE <sub>01</sub> |                       |                      |                       |                      |
| TE <sub>11</sub> |                       |                      |                       |                      |
| TE <sub>20</sub> |                       |                      |                       |                      |
| TE <sub>21</sub> |                       |                      |                       |                      |
| TE <sub>02</sub> |                       |                      |                       |                      |

- Result:

Five TE modes can propagate TE<sub>10</sub>, TE<sub>01</sub>, TE<sub>11</sub>, TE<sub>20</sub>, TE<sub>21</sub>, TE<sub>02</sub>

### 2. Output for TE<sub>10</sub> Mode:

Using the program with a=5 m b=2.5 m and f=10 GHz:

| Parameter                  | Simulation Results | Theoretical Results |
|----------------------------|--------------------|---------------------|
|                            | Value              | Value               |
| Cutoff Frequency ( $f_c$ ) |                    |                     |
| Phase Velocity ( $v_p$ )   |                    |                     |
| Group Velocity ( $v_g$ )   |                    |                     |

### RESULT:

The group velocity, phase velocity and phase constant of waves in a rectangular waveguide are estimated using SCILAB.

#### **Test Case 4 :**

**A research lab is experimenting with high-order TE modes ( $m,n>1$ ) in a waveguide with dimensions  $a=3.0$  cm  $b=1.5$  cm, and an operating frequency of  $f=12$  GHz.**

- 1. Calculate the cutoff frequencies for  $TE_{10}$  mode.**
- 2. Compute the phase and group velocities of this mode.**
- 3. Discuss how high-order modes can be utilized for advanced applications such as mode multiplexing in communications.**

#### **AIM:**

To find the group velocity, phase velocity and phase constant of Electromagnetic waves in Rectangular Waveguides whose dimensions are  $a=3.0$  cm,  $b=1.5$  cm and the operating frequency is  $f=12$  GHz.

#### **SOFTWARE REQUIRED:**

SCILAB Version 6.1.1

#### **PROCEDURE:**

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

#### **PROGRAM:**

```
clc;
clear;
close;
//TE10 Mode
a=8.636*10^-2;
b=4.318*10^-2;
f=4*10^9;
V=3*10^8;
m=input("enter the value of m= ");
```

```

n=input("enter the value of n= ");
fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);
disp("Cut off frequency=",fc*10^-9);
//let phase velocity = A
A=V/sqrt(1-(fc/f)^2);
disp("Phase velocity in Mm/sec=",A*10^-6);
//let group velocity = B
B=V*A/A;
disp("Group velocity in Mm/sec =", B*10^-6);

```

## OUTPUT:

### *Waveguide Parameters:*

- Width (a) = 3.0 cm=0.03 m
- Height (b) = 1.5 cm=0.015 m
- Operating Frequency (f) = 12 GHz= $12 \times 10^9$  Hz
- Speed of Light (V) =  $3 \times 10^8$  m/s

### 1. No. of Modes:

To analyze all modes, calculate  $f_c$  for other modes with  $m, n > 1$ .

## OUTPUT:

|                  | Simulation Results    |                      | Theoretical Results   |                      |
|------------------|-----------------------|----------------------|-----------------------|----------------------|
| Mode (m,n)       | Cutoff Frequency (fc) | Propagation (Yes/No) | Cutoff Frequency (fc) | Propagation (Yes/No) |
| TE <sub>10</sub> |                       |                      |                       |                      |
| TE <sub>20</sub> |                       |                      |                       |                      |
| TE <sub>01</sub> |                       |                      |                       |                      |
| TE <sub>11</sub> |                       |                      |                       |                      |
| TE <sub>21</sub> |                       |                      |                       |                      |
| TE <sub>30</sub> |                       |                      |                       |                      |

### 2. Program Output for TE<sub>10</sub> Mode:

Using the program with a=0.03 m b=0.015 m and f=12 GHz:

| Parameter                  | Simulation Results | Theoretical Results |
|----------------------------|--------------------|---------------------|
|                            | Value              | Value               |
| Cutoff Frequency ( $f_c$ ) |                    |                     |
| Phase Velocity ( $v_p$ )   |                    |                     |
| Group Velocity ( $v_g$ )   |                    |                     |

## RESULT:

The group velocity, phase velocity and phase constant of waves in a rectangular waveguide are estimated using SCILAB.

|                |                                                                                                                                                                                                                           |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EXNO: 6</b> | <b>Inference of Q Factor of Rectangular Cavity Resonator for TE<sub>101</sub> mode with the following inputs</b><br><b>a) Dimension a in x axis</b><br><b>b) Dimension b in y axis</b><br><b>c) Dimension p in z axis</b> |
| <b>DATE:</b>   |                                                                                                                                                                                                                           |

#### Test Case 1:

You are tasked with designing a rectangular cavity resonator that operates in the TE<sub>101</sub> mode. The dimensions of the resonator are a=5 cm, b=4 cm, and c=10 cm. The cavity is made of copper (C=5.8×10<sup>7</sup> S/m).

1. If the dimension a is increased to 7 cm while keeping b and c constant, how does the Q-factor change? Explain the relationship between a and the Q-factor.

**Hint:** Use the given formula for Q and compare the resulting values.

#### AIM:

To determine the Q Factor of Rectangular Cavity Resonator for TE<sub>101</sub> mode by varying three dimensions

#### SOFTWARE REQUIRED:

SCILAB

#### PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

#### CALCULATION

$$Q_{101} = \frac{\pi f_{101} \mu_0 a b d (a^2 + d^2)}{R_s [2b(a^3 + d^3) + ad(a^2 + d^2)]} \quad (\text{TE}_{101} \text{ mode}),$$

#### PROGRAM:

```
clear;
clc;
a=5*10^-2,b=4*10^-2,c=10*10^-2,C=5.8*10^7,Uo=4*%pi*10^-7;
f101=3.335*10^9;
```

```

d=sqrt(1/(%pi*f101*Uo*C));
Q=(a*a+c*c)*a*b*c/(d*(2*b*(a^3 + c^3)+a*c*(a*a+c*c)));
disp(Q,'Quality factor of TE101 = ');

```

### OUTPUT:

| a (cm) | b (cm) | c (cm) | Simulation Results | Theoretical Results |
|--------|--------|--------|--------------------|---------------------|
|        |        |        | Q Factor           | Q Factor            |
| 4.0    | 3.0    | 8.0    |                    |                     |
| 4.0    | 3.0    | 9.0    |                    |                     |
| 4.0    | 3.0    | 10.0   |                    |                     |
| 4.0    | 3.0    | 11.0   |                    |                     |
| 4.0    | 3.0    | 12.0   |                    |                     |

**RESULT:** The Q Factor of Rectangular Cavity Resonator for TE<sub>101</sub> mod by varying three dimensions is determined using Scilab

### Test Case 2 :

You are asked to optimize the rectangular cavity resonator to achieve the highest possible Q-factor using the dimensions provided.

1. Based on the variation in a, b, and c, which combination of dimensions yields the maximum Q-factor?
2. Explain why this combination results in a higher Q-factor. Consider the relationship between a, b, and c and how they affect the Q-factor in the given formula.

### AIM:

To determine the Q Factor of Rectangular Cavity Resonator for TE<sub>101</sub> mode by varying three dimensions

### SOFTWARE REQUIRED:

SCILAB

## PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

## CALCULATION

$$Q_{101} = \frac{\pi f_{101} \mu_0 a b d (a^2 + d^2)}{R_s [2b(a^3 + d^3) + ad(a^2 + d^2)]} \quad (\text{TE}_{101} \text{ mode}),$$

## PROGRAM:

```
clear;
clc;
a=5*10^-2, b=4*10^-2, c=10*10^-2, C=5.8*10^7, Uo=4*%pi*10^-7;
f101=3.335*10^9;
d=sqrt(1/(%pi*f101*Uo*C));
Q=(a*a+c*c)*a*b*c/(d*(2*b*(a^3 + c^3)+a*c*(a*a+c*c)));
disp(Q,'Quality factor of TE101 = ');
```

## OUTPUT:

The combination of dimensions that yields the highest Q-factor is as follows:

| a (cm) | b (cm) | c (cm) | Simulation Results | Theoretical Results |
|--------|--------|--------|--------------------|---------------------|
|        |        |        | Q Factor           | Q Factor            |
|        |        |        |                    |                     |

## RESULT:

The Q Factor of Rectangular Cavity Resonator for TE101 mode by varying three dimensions is determined using Scilab.

### Test Case 3 :

Your task is to assess how changes in a single dimension affect the resonator's performance.

1. Fix  $b=4$  cm and  $c=10$  cm and vary  $a$  between 4 cm and 6 cm. How does the Q-factor change? What does this tell you about the sensitivity of the Q-factor to the dimension  $a$ ?
2. In practical scenarios, which dimension is most critical to optimize for achieving a high Q-factor, and why?

### AIM:

To determine the Q Factor of Rectangular Cavity Resonator for  $TE_{101}$  mode by varying three dimensions

### SOFTWARE REQUIRED:

SCILAB

### PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

### CALCULATION

$$Q_{101} = \frac{\pi f_{101} \mu_0 a b d (a^2 + d^2)}{R_s [2b(a^3 + d^3) + ad(a^2 + d^2)]} \quad (TE_{101} \text{ mode}),$$

### PROGRAM:

```
clear;
clc;
a=5*10^-2, b=4*10^-2, c=10*10^-2, C=5.8*10^7, Uo=4*pi*10^-7;
f101=3.335*10^9;
d=sqrt(1/(pi*f101*Uo*C));
Q=(a*a+c*c)*a*b*c/(d*(2*b*(a^3 + c^3)+a*c*(a*a+c*c)));
disp(Q,'Quality factor of TE101 = ');
```

## OUTPUT:

Below is the result of how changes in the dimension of 'a' affect the Q-factor, we can calculate the Q-factor for fixed values of  $b=4$  cm and  $c=10$  cm, while varying 'a' between 4 cm and 6 cm in equal steps.

### 1. Q-Factor Calculation for Varying a:

| a (cm) | b (cm) | c (cm) | Simulation Results | Theoretical Results |
|--------|--------|--------|--------------------|---------------------|
|        |        |        | Q-Factor           | Q-Factor            |
| 4.0    | 4.0    | 10.0   |                    |                     |
| 4.5    | 4.0    | 10.0   |                    |                     |
| 5.0    | 4.0    | 10.0   |                    |                     |
| 5.5    | 4.0    | 10.0   |                    |                     |
| 6.0    | 4.0    | 10.0   |                    |                     |

## RESULT:

The Q Factor of Rectangular Cavity Resonator for TE<sub>101</sub> mode by varying three dimensions is determined using Scilab.

### Test Case 4 :

**You are tasked with investigating how changes in b(the y-axis dimension) affect the Q-factor of the rectangular cavity resonator. Fix  $a=5$  cm and  $c=10$  cm and vary b between 3 cm and 5 cm. Analyze the Q-factor for these changes.**

### AIM:

To determine the Q Factor of Rectangular Cavity Resonator for TE<sub>101</sub> mode by varying three dimensions

### SOFTWARE REQUIRED:

SCILAB

### PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values

- Type the program
- Execute the program
- End the program

### CALCULATION

$$Q_{101} = \frac{\pi f_{101} \mu_0 a b d (a^2 + d^2)}{R_s [2b(a^3 + d^3) + ad(a^2 + d^2)]} \quad (\text{TE}_{101} \text{ mode}),$$

### PROGRAM:

```
clear;
clc;
a=5*10^-2, b=4*10^-2, c=10*10^-2, C=5.8*10^7, Uo=4*%pi*10^-7;
f101=3.335*10^9;
d=sqrt(1/(%pi*f101*Uo*C));
Q=(a*a+c*c)*a*b*c/(d*(2*b*(a^3 + c^3)+a*c*(a*a+c*c)));
disp(Q,'Quality factor of TE101 = ');
```

### OUTPUT:

#### Calculation of Q-Factor for Varying b

| a (cm) | b (cm) | c (cm) | Simulation Results | Theoretical Results |
|--------|--------|--------|--------------------|---------------------|
|        |        |        | Q-Factor           | Q-Factor            |
| 5.0    | 3.0    | 10.0   |                    |                     |
| 5.0    | 3.5    | 10.0   |                    |                     |
| 5.0    | 4.0    | 10.0   |                    |                     |
| 5.0    | 4.5    | 10.0   |                    |                     |
| 5.0    | 5.0    | 10.0   |                    |                     |

### RESULT:

The Q Factor of Rectangular Cavity Resonator for TE101 mode by varying three dimensions is determined using Scilab.

|                |                                                                                                                                                               |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EXNO: 7</b> | <b>Design a two conductor transmission line and Investigate the impact of Dielectric layer thickness on characteristic Impedance and Propagation Velocity</b> |
| <b>DATE:</b>   |                                                                                                                                                               |

**Test Case 1: Design a Two conductor transmission line with single ground plane and estimate the impact of varying the thickness of the dielectric layer on characteristic Impedance and Propagation Velocity. Thickness to be varied between 5 to 15mm,**

**AIM:** To design a Two conductor transmission line with single ground plane and estimate the impact of varying the thickness of the dielectric layer on characteristic Impedance and Propagation Velocity. Thickness to be varied between 5 to 15mm,

#### **TOOL REQUIRED:**

#### **TNT 1.2.2**

#### **PROCEDURE:**

- **Open TNT 1.2.2 Tool**
- **Open file-new**
- **Create a two conductor transmission line**
- **Select the dielectric layer with specified material and dimensions**
- **Select the ground plane with specified dimensions**
- **Place the dielectric layer on top of the ground plane**
- **Select two rectangular conductors with specified material and dimensions and place on the top of dielectric layer.**

#### **CALCULATION**

$$Z_0 = \frac{276}{\sqrt{k}} \log \frac{d}{r}$$

$Z_0$  = Characteristic impedance of line

$d$  = Distance between conductor centers

$r$  = Conductor radius

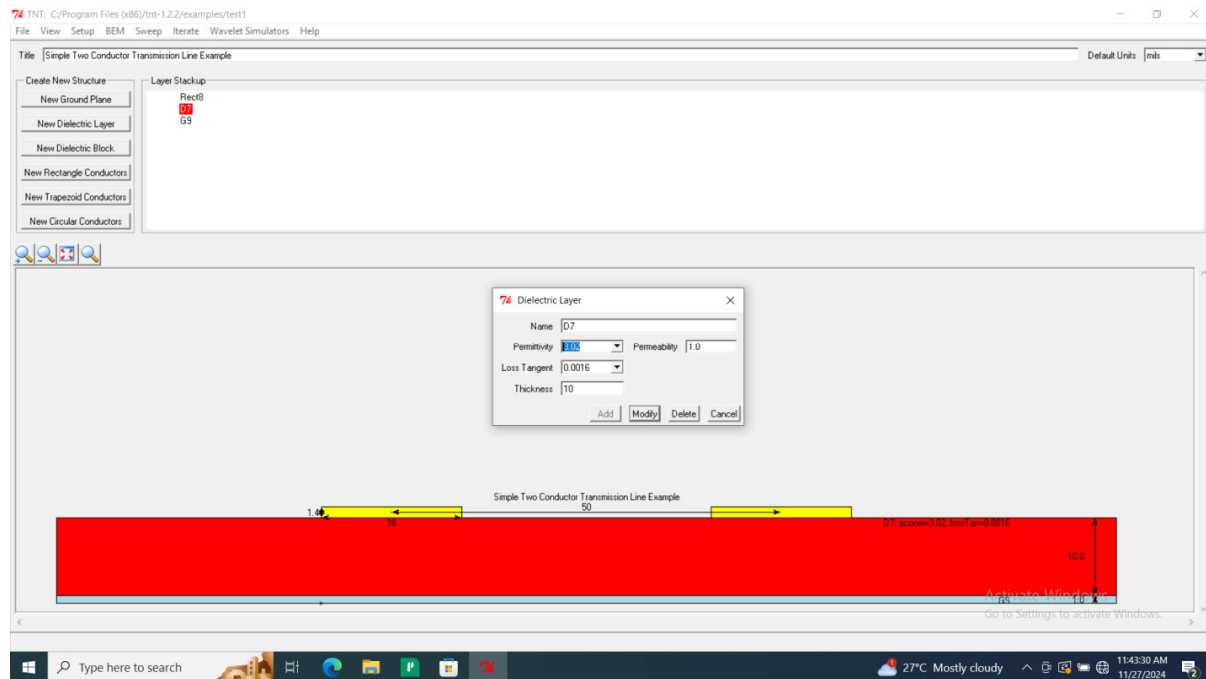
$k$  = Relative permittivity of insulation between conductors

$$\text{Velocity factor} = \frac{v}{c} = \frac{1}{\sqrt{k}}$$

$v$  = Velocity of wave propagation

$c$  = Velocity of light in a vacuum

$k$  = Relative permittivity of insulation between conductors



## OUTPUT:

| Thickness of Dielectric layer (mm) | Simulation Results              |                            | Theoretical Results             |                            |
|------------------------------------|---------------------------------|----------------------------|---------------------------------|----------------------------|
|                                    | Characteristic Impedance (ohms) | Propagation Velocity (m/s) | Characteristic Impedance (ohms) | Propagation Velocity (m/s) |
| 5                                  |                                 |                            |                                 |                            |
| 10                                 |                                 |                            |                                 |                            |
| 12                                 |                                 |                            |                                 |                            |
| 15                                 |                                 |                            |                                 |                            |

## RESULT:

A Two conductor transmission line is designed with single ground plane and the impact of varying the thickness of the dielectric layer on characteristic Impedance and Propagation Velocity is estimated.

**Test Case 2: Design a Two conductor transmission line with single ground plane and investigate the effect of far end cross talk and near end cross talk by varying the thickness of the dielectric layer .Thickness to be varied between 5 to 15mm**

**AIM:** To design a Two conductor transmission line with single ground plane and estimate the impact of varying the thickness of the dielectric layer on characteristic Impedance and Propagation Velocity. Thickness to be varied between 5 to 15mm,

## TOOL REQUIRED:

TNT 1.2.2

## PROCEDURE:

- Open TNT 1.2.2 Tool
- Open file-new
- Create a two conductor transmission line
- Select the dielectric layer with specified material and dimensions
- Select the ground plane with specified dimensions
- Place the dielectric layer on top of the ground plane
- Select two rectangular conductors with specified material and dimensions and place on the top of dielectric layer.

## CALCULATION

$$Z_0 = \frac{276}{\sqrt{k}} \log \frac{d}{r}$$

$Z_0$  = Characteristic impedance of line

$d$  = Distance between conductor centers

$r$  = Conductor radius

$k$  = Relative permittivity of insulation between conductors

$$\text{Velocity factor} = \frac{v}{c} = \frac{1}{\sqrt{k}}$$

v = Velocity of wave propagation

c = Velocity of light in a vacuum

k = Relative permittivity of insulation between conductors

### OUTPUT:

| Thickness of Dielectric layer | Simulation Results      |                          | Theoretical Results     |                          |
|-------------------------------|-------------------------|--------------------------|-------------------------|--------------------------|
|                               | Far end cross talk (dB) | Near end cross talk (dB) | Far end cross talk (dB) | Near end cross talk (dB) |
| 5                             |                         |                          |                         |                          |
| 10                            |                         |                          |                         |                          |
| 12                            |                         |                          |                         |                          |
| 15                            |                         |                          |                         |                          |

### RESULT:

A Two conductor transmission line is designed with single ground plane and the effect of far end cross talk and near end cross talk is investigated by varying the thickness of the dielectric layer .

|                |                                                                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EXNO: 8</b> | <b>Design a two conductor transmission line and examine the impact of conductor dimensions on characteristic Impedance and Propagation Velocity</b> |
| <b>DATE:</b>   |                                                                                                                                                     |

**Test Case 1: Design a Two conductor transmission line with single ground plane and investigate the impact of the width of the rectangular conductor on Characteristic Impedance and Propagation Velocity .Thickness to be varied between 10 to 20 mm.**

**AIM:** To design a Two conductor transmission line with single ground plane and investigate the impact of the width of the rectangular conductor on Characteristic Impedance and Propagation Velocity .Thickness to be varied between 10 to 20 mm.

### **TOOL REQUIRED:**

TNT 1.2.2

### **PROCEDURE:**

- Open TNT 1.2.2 Tool
- Open file-new
- Create a two conductor transmission line
- Select the dielectric layer with specified material and dimensions
- Select the ground plane with specified dimensions
- Place the dielectric layer on top of the ground plane
- Select two rectangular conductors with specified material and dimensions and place on the top of dielectric layer.

### **CALCULATION**

$$Z_0 = \frac{276}{\sqrt{k}} \log \frac{d}{r}$$

$Z_0$  = Characteristic impedance of line

$d$  = Distance between conductor centers

$r$  = Conductor radius

$k$  = Relative permittivity of insulation between conductors

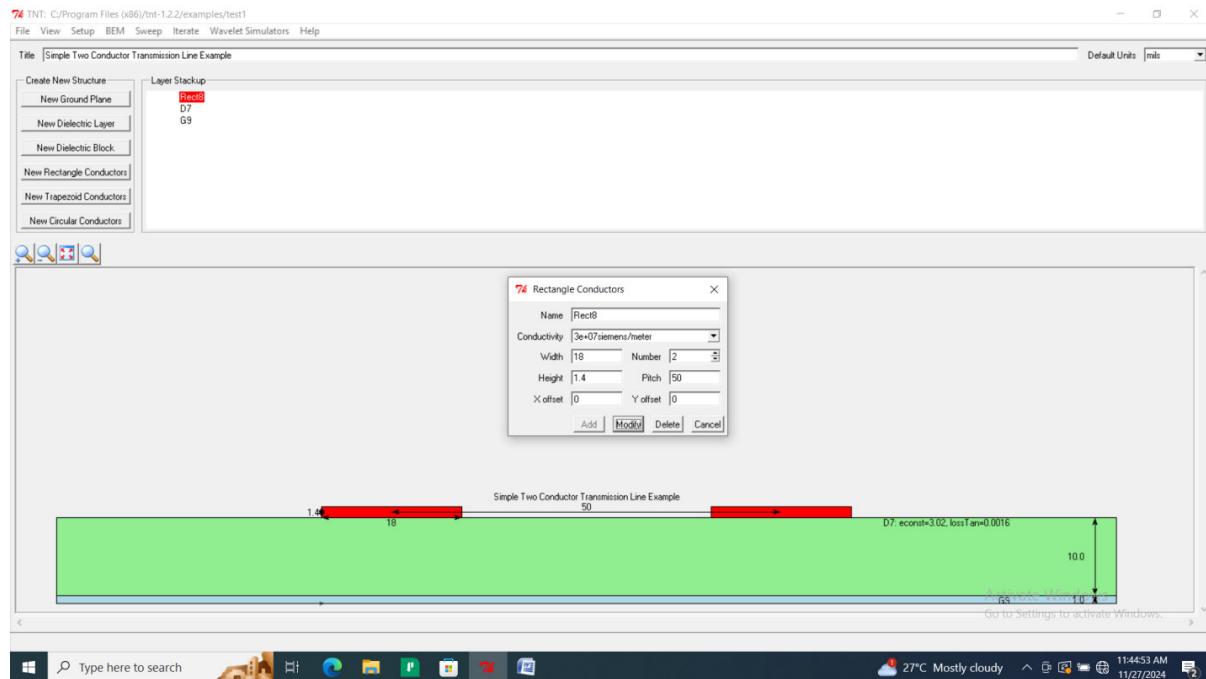
$$\text{Velocity factor} = \frac{v}{c} = \frac{1}{\sqrt{k}}$$

$v$  = Velocity of wave propagation

$c$  = Velocity of light in a vacuum

$k$  = Relative permittivity of insulation between conductors

## OUTPUT:



| Width of the Conductor (mm) | Simulation Results              |                            | Theoretical Results             |                            |
|-----------------------------|---------------------------------|----------------------------|---------------------------------|----------------------------|
|                             | Characteristic Impedance (ohms) | Propagation Velocity (m/s) | Characteristic Impedance (ohms) | Propagation Velocity (m/s) |
| 20                          |                                 |                            |                                 |                            |
| 18                          |                                 |                            |                                 |                            |
| 15                          |                                 |                            |                                 |                            |
| 10                          |                                 |                            |                                 |                            |

## RESULT:

A Two conductor transmission line with single ground plane is designed and the impact of the width of the rectangular conductor on Characteristic Impedance and Propagation Velocity is investigated.

**Test Case 2: Examine the effect of Characteristic Impedance and Propagation Velocity by varying the height of the rectangular conductor .Height to be varied between 1 to 1.4 mm.**

**AIM:** To examine the effect of Characteristic Impedance and Propagation Velocity by varying the height of the rectangular conductor .Height to be varied between 1 to 1.4 mm.

## TOOL REQUIRED:

TNT 1.2.2

## PROCEDURE:

- Open TNT 1.2.2 Tool
- Open file-new
- Create a two conductor transmission line
- Select the dielectric layer with specified material and dimensions
- Select the ground plane with specified dimensions
- Place the dielectric layer on top of the ground plane
- Select two rectangular conductors with specified material and dimensions and place on the top of dielectric layer.

## CALCULATION

$$Z_0 = \frac{276}{\sqrt{k}} \log \frac{d}{r}$$

$Z_0$  = Characteristic impedance of line

$d$  = Distance between conductor centers

$r$  = Conductor radius

$k$  = Relative permittivity of insulation between conductors

$$\text{Velocity factor} = \frac{v}{c} = \frac{1}{\sqrt{k}}$$

v = Velocity of wave propagation

c = Velocity of light in a vacuum

k = Relative permittivity of insulation between conductors

### OUTPUT:

| Height of the Conductor (mm) | Simulation Results              |                            | Theoretical Results             |                            |
|------------------------------|---------------------------------|----------------------------|---------------------------------|----------------------------|
|                              | Characteristic Impedance (ohms) | Propagation Velocity (m/s) | Characteristic Impedance (ohms) | Propagation Velocity (m/s) |
| 1                            |                                 |                            |                                 |                            |
| 1.2                          |                                 |                            |                                 |                            |
| 1.3                          |                                 |                            |                                 |                            |
| 1.4                          |                                 |                            |                                 |                            |

### RESULT:

The effect of Characteristic Impedance and Propagation Velocity is examined by varying the height of the rectangular conductor between 1 to 1.4 mm.

|                |                                                                                                                                                          |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EXNO: 9</b> | <b>Design a two conductor transmission line and examine the impact of Dielectric layer material on characteristic Impedance and Propagation Velocity</b> |
| <b>DATE:</b>   |                                                                                                                                                          |

**Test Case 1: Design a Two conductor transmission line with single ground plane and investigate the impact of the Dielectric layer material on characteristic Impedance and Propagation Velocity. Materials to be tested - GETEK ,FR 404, ADS995 and RO3203 .**

**AIM:** To design a Two conductor transmission line with single ground plane and investigate the impact of the Dielectric layer material on characteristic Impedance and Propagation Velocity. Materials to be tested - GETEK ,FR 404, ADS995 and RO3203 .

### **TOOL REQUIRED:**

TNT 1.2.2

### **PROCEDURE:**

- Open TNT 1.2.2 Tool
- Open file-new
- Create a two conductor transmission line
- Select the dielectric layer with specified material and dimensions
- Select the ground plane with specified dimensions
- Place the dielectric layer on top of the ground plane
- Select two rectangular conductors with specified material and dimensions and place on the top of dielectric layer.

### **CALCULATION**

$$Z_0 = \frac{276}{\sqrt{k}} \log \frac{d}{r}$$

$Z_0$  = Characteristic impedance of line

$d$  = Distance between conductor centers

$r$  = Conductor radius

$k$  = Relative permittivity of insulation between conductors

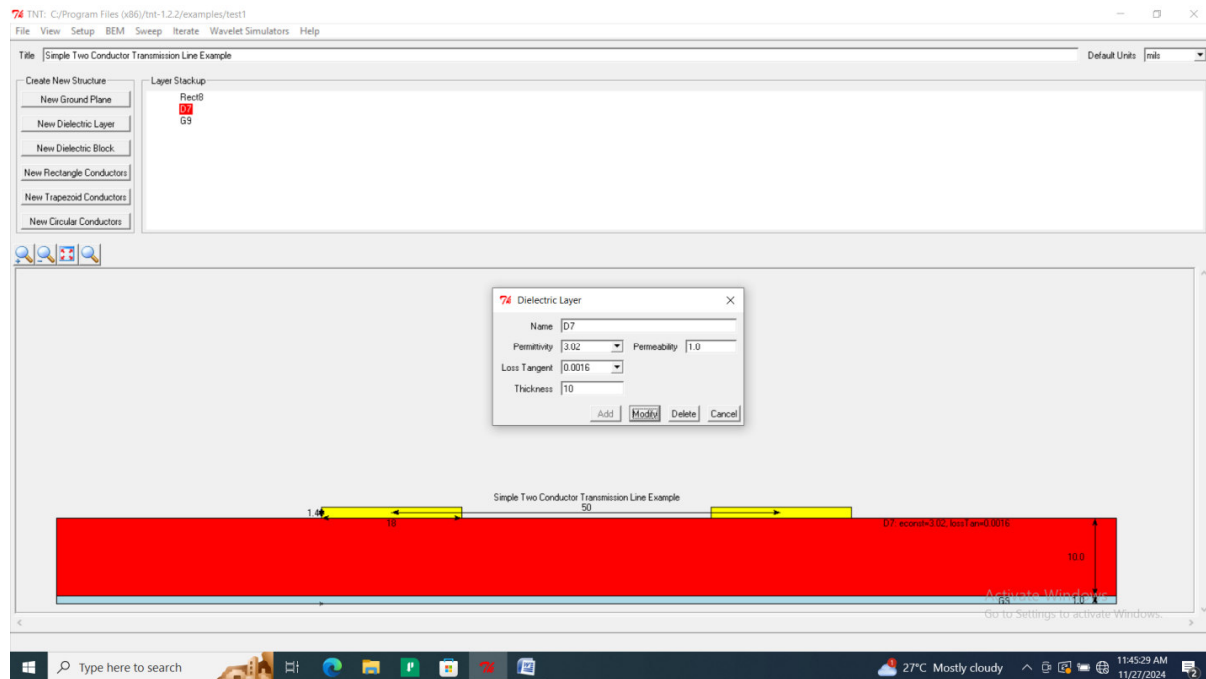
$$\text{Velocity factor} = \frac{v}{c} = \frac{1}{\sqrt{k}}$$

$v$  = Velocity of wave propagation

$c$  = Velocity of light in a vacuum

$k$  = Relative permittivity of insulation between conductors

## OUTPUT:



| Dielectric layer material | Permittivity | Loss tangent | Simulation Results              |                            | Theoretical Results             |                            |
|---------------------------|--------------|--------------|---------------------------------|----------------------------|---------------------------------|----------------------------|
|                           |              |              | Characteristic Impedance (ohms) | Propagation Velocity (m/s) | Characteristic Impedance (ohms) | Propagation Velocity (m/s) |
| GETEK                     |              |              |                                 |                            |                                 |                            |
| FR 404                    |              |              |                                 |                            |                                 |                            |
| ADS995                    |              |              |                                 |                            |                                 |                            |
| RO3203                    |              |              |                                 |                            |                                 |                            |

## RESULT:

A Two conductor transmission line with single ground plane is constructed and the impact of the Dielectric layer material like GETEK, FR 404, ADS995 and RO3203 on characteristic

Impedance and Propagation Velocity is examined. Materials to be tested -

**Test Case 2: Design a Two conductor transmission line with single ground plane and investigate the impact of the conductor material on Rdc parameters. Width of the conductor is 18 mm and the height is 1.4 mm. Conductor is placed on a 10 mm thickness dielectric layer.**

**AIM:** To design a Two conductor transmission line with single ground plane and investigate the impact of the conductor material on Rdc parameters. Width of the conductor is 18 mm and the height is 1.4 mm. Conductor is placed on a 10 mm thickness dielectric layer.

### **TOOL REQUIRED:**

TNT 1.2.2

### **PROCEDURE:**

- Open TNT 1.2.2 Tool
- Open file-new
- Create a two conductor transmission line
- Select the dielectric layer with specified material and dimensions
- Select the ground plane with specified dimensions
- Place the dielectric layer on top of the ground plane
- Select two rectangular conductors with specified material and dimensions and place on the top of dielectric layer.

### **CALCULATION**

$$Z_0 = \frac{276}{\sqrt{k}} \log \frac{d}{r}$$

$Z_0$  = Characteristic impedance of line

$d$  = Distance between conductor centers

$r$  = Conductor radius

$k$  = Relative permittivity of insulation between conductors

$$\text{Velocity factor} = \frac{v}{c} = \frac{1}{\sqrt{k}}$$

v = Velocity of wave propagation

c = Velocity of light in a vacuum

k = Relative permittivity of insulation between conductors

### OUTPUT:

| Conductor material | Conductivity S/m | Rdc R1 R1 | Rdc R1 R0 | Rdc R0 R1 | Rdc R0 R0 |
|--------------------|------------------|-----------|-----------|-----------|-----------|
| <b>Tungston</b>    |                  |           |           |           |           |
| <b>Aluminium</b>   |                  |           |           |           |           |
| <b>Copper</b>      |                  |           |           |           |           |
| <b>Silver</b>      |                  |           |           |           |           |

### RESULT:

A Two conductor transmission line with single ground plane is designed and the impact of the conductor material on Rdc parameters is evaluated.

|                 |                                                                                                                                                 |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EXNO: 10</b> | <b>Simulation and Analysis of a Coaxial Connector Using CST Studio Suite and examining the power distribution among ports and S-parameters.</b> |
| <b>DATE:</b>    |                                                                                                                                                 |

### **Test Case 1:**

To simulate a coaxial connector using CST Studio Suite, analyze its electromagnetic behavior, and examine the power distribution among ports over a specified frequency range.

**AIM:** To simulate a coaxial connector using CST Studio Suite, analyze its electromagnetic behavior, and examine the power distribution among ports over a specified frequency range.

### **TOOL REQUIRED:**

CST Studio Suite

### **PROCEDURE:**

Open CST Studio Suite

Open file-new

Create a two port coaxial connector

Define ports

Ensure proper port mode selection (TEM mode).

Run the simulation.

Observe:

- S11 (Reflection Coefficient)
- S21 (Transmission Coefficient)
- Power accepted at Port 1
- Power transmitted to Port 2

Plot: Power distribution at ports

### **CALCULATION:**

$$P_{reflected} = |S_{11}|^2 P_{input}$$

$$P_{transmitted} = |S_{21}|^2 P_{input}$$

$$P_{loss} = P_{input} - (P_{reflected} + P_{transmitted})$$

Two-Port Network S-Parameter Matrix

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

Input Reflection Coefficient –  $S_{11}$

$$S_{11} = \left. \frac{b_1}{a_1} \right|_{a_2=0}$$

Forward Transmission Coefficient –  $S_{21}$

$$S_{21} = \left. \frac{b_2}{a_1} \right|_{a_2=0}$$

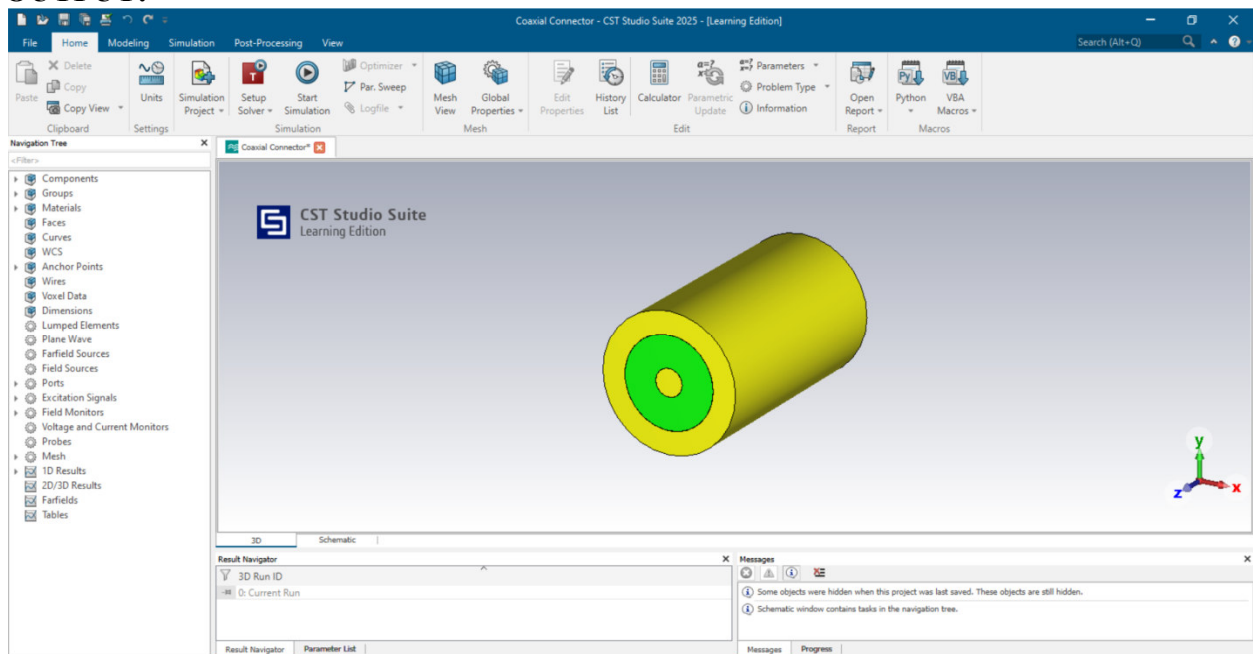
Reverse Transmission Coefficient –  $S_{12}$

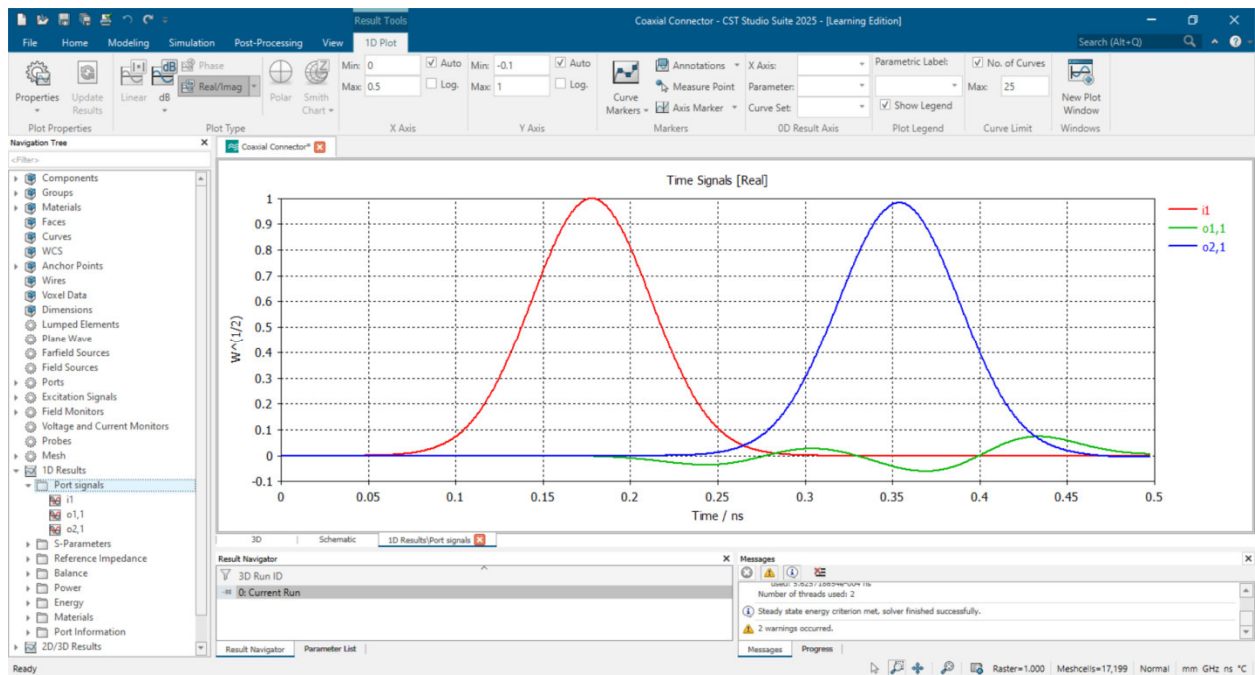
$$S_{12} = \left. \frac{b_1}{a_2} \right|_{a_1=0}$$

Output Reflection Coefficient –  $S_{22}$

$$S_{22} = \left. \frac{b_2}{a_2} \right|_{a_1=0}$$

**OUTPUT:**





| Port (i-input,<br>o-output) | Time (ns) | Power $W^{1/2}$ |
|-----------------------------|-----------|-----------------|
| i1                          |           |                 |
| o1,1                        |           |                 |
| o2,1                        |           |                 |

## RESULT:

The coaxial connector was successfully simulated using CST Studio Suite over the specified frequency range. The electromagnetic behavior of the structure was analyzed through power distribution among ports shows that the majority of the input power is transmitted to the output port with minimal reflection and negligible losses, indicating efficient signal transmission and satisfactory connector performance.

## Test Case 2:

To simulate a coaxial connector using CST Studio Suite, analyze its electromagnetic behavior, and examine the power distribution among ports over a specified frequency range.

**AIM:** To simulate a coaxial connector using CST Studio Suite, analyze its electromagnetic behavior, and examine the power distribution among ports over a specified frequency range.

## TOOL REQUIRED:

CST Studio Suite

## PROCEDURE:

Open CST Studio Suite

Open file-new

Create a two port coaxial connector

Define ports

Ensure proper port mode selection (TEM mode).

Run the simulation.

Observe:

- S11 (Reflection Coefficient)
- S21 (Transmission Coefficient)
- Power accepted at Port 1
- Power transmitted to Port 2

Plot: S-parameter vs Frequency

## CALCULATION

Two-Port Network S-Parameter Matrix

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

Input Reflection Coefficient – S<sub>11</sub>

$$S_{11} = \left. \frac{b_1}{a_1} \right|_{a_2=0}$$

Forward Transmission Coefficient – S<sub>21</sub>

$$S_{21} = \left. \frac{b_2}{a_1} \right|_{a_2=0}$$

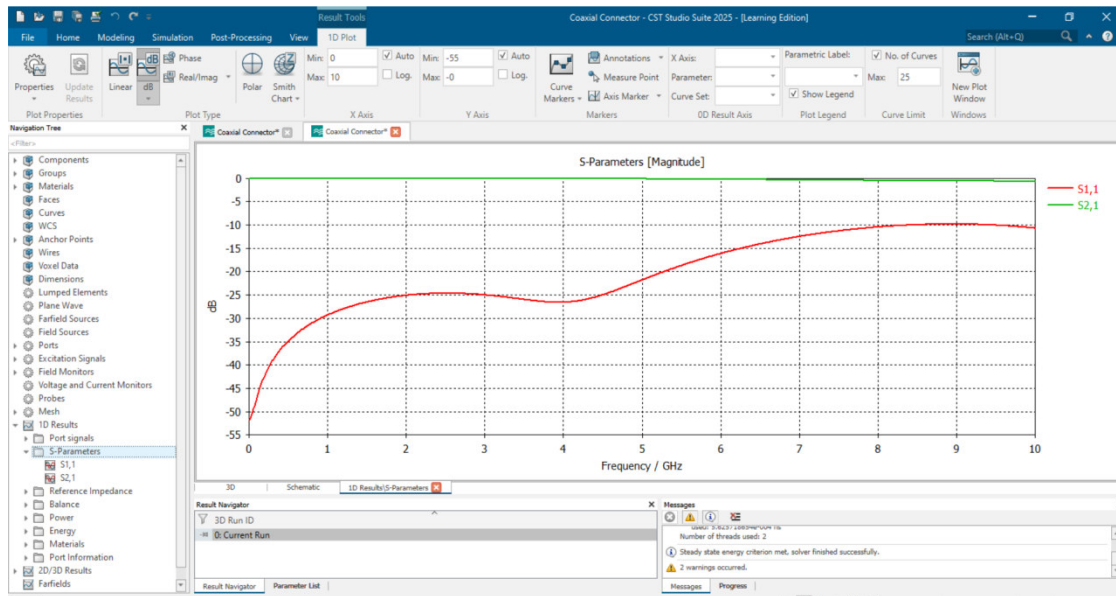
Reverse Transmission Coefficient – S<sub>12</sub>

$$S_{12} = \left. \frac{b_1}{a_2} \right|_{a_1=0}$$

## Output Reflection Coefficient – $S_{22}$

$$S_{22} = \frac{b_2}{a_2} \bigg|_{a_1=0}$$

## OUTPUT:



| Frequency (GHz) | S Parameters Magnitude (dB) |
|-----------------|-----------------------------|
| S11             |                             |
| S21             |                             |

## RESULT:

The coaxial connector was successfully simulated using CST Studio Suite over the specified frequency range. The electromagnetic behavior of the structure was analyzed through S parameters that show that the majority of the input power is transmitted to the output port with minimal reflection and negligible losses, indicating efficient signal transmission and satisfactory connector performance.

|                 |                                                                                                                                        |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>EXNO: 11</b> | <b>Electromagnetic Field and Power Flow Analysis of a Rectangular Waveguide in CST Studio Suite and determination of S parameters.</b> |
| <b>DATE:</b>    |                                                                                                                                        |

### **Test Case 1:**

To simulate a Rectangular Waveguide using CST Studio Suite, analyze its electromagnetic behavior, and examine the power distribution among ports over a specified frequency range.

**AIM:** To simulate a Rectangular Waveguide using CST Studio Suite, analyze its electromagnetic behavior, and examine the power distribution among ports over a specified frequency range.

### **TOOL REQUIRED:**

CST Studio Suite

### **PROCEDURE:**

Open CST Studio Suite

Open file-new

Create a rectangular waveguide section

Define ports

Run the simulation.

Observe:

- S11 (Reflection Coefficient)
- S21 (Transmission Coefficient)
- Power accepted at Port 1
- Power transmitted to Port 2

Plot the Power distribution at ports

### **CALCULATION**

$$P_{reflected} = |S_{11}|^2 P_{input}$$

$$P_{transmitted} = |S_{21}|^2 P_{input}$$

$$P_{loss} = P_{input} - (P_{reflected} + P_{transmitted})$$

## Two-Port Network S-Parameter Matrix

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

Input Reflection Coefficient –  $S_{11}$

$$S_{11} = \left. \frac{b_1}{a_1} \right|_{a_2=0}$$

Forward Transmission Coefficient –  $S_{21}$

$$S_{21} = \left. \frac{b_2}{a_1} \right|_{a_2=0}$$

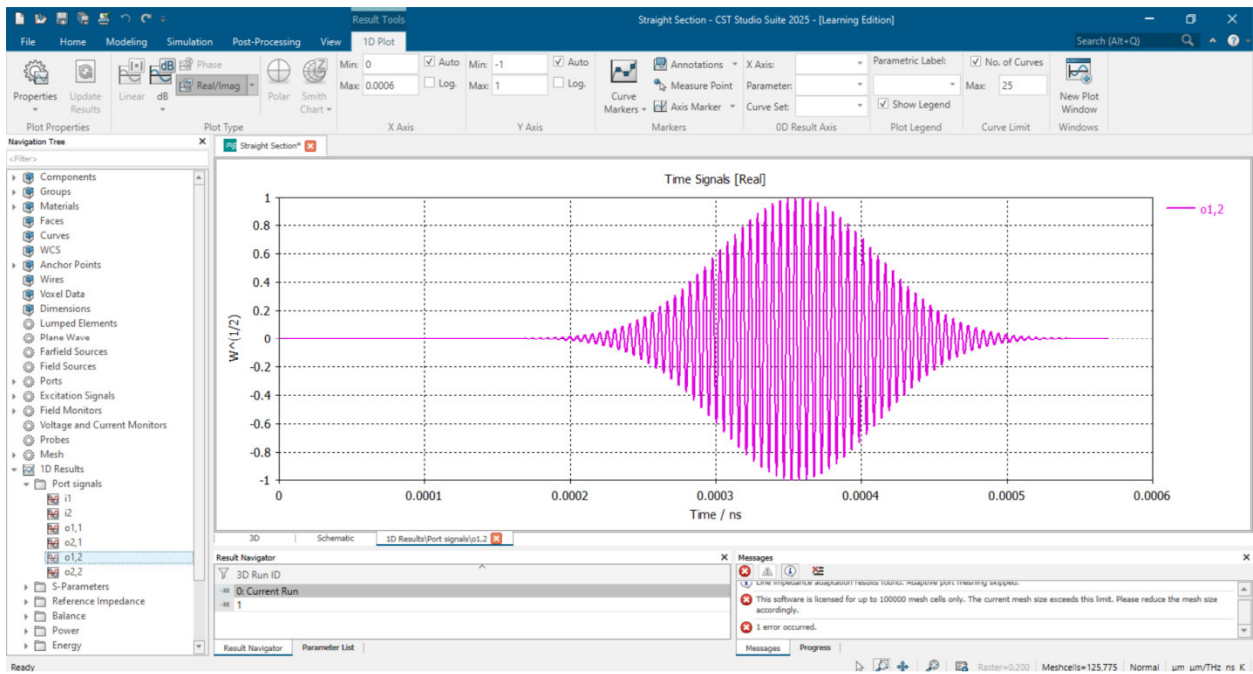
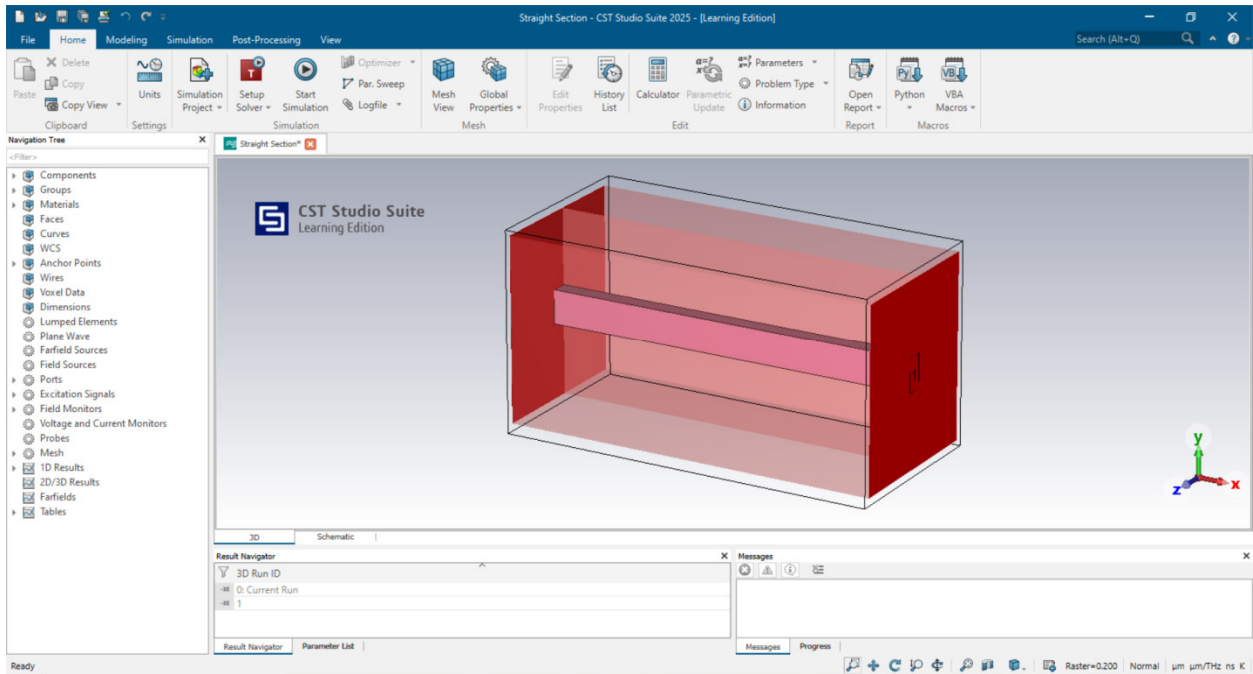
Reverse Transmission Coefficient –  $S_{12}$

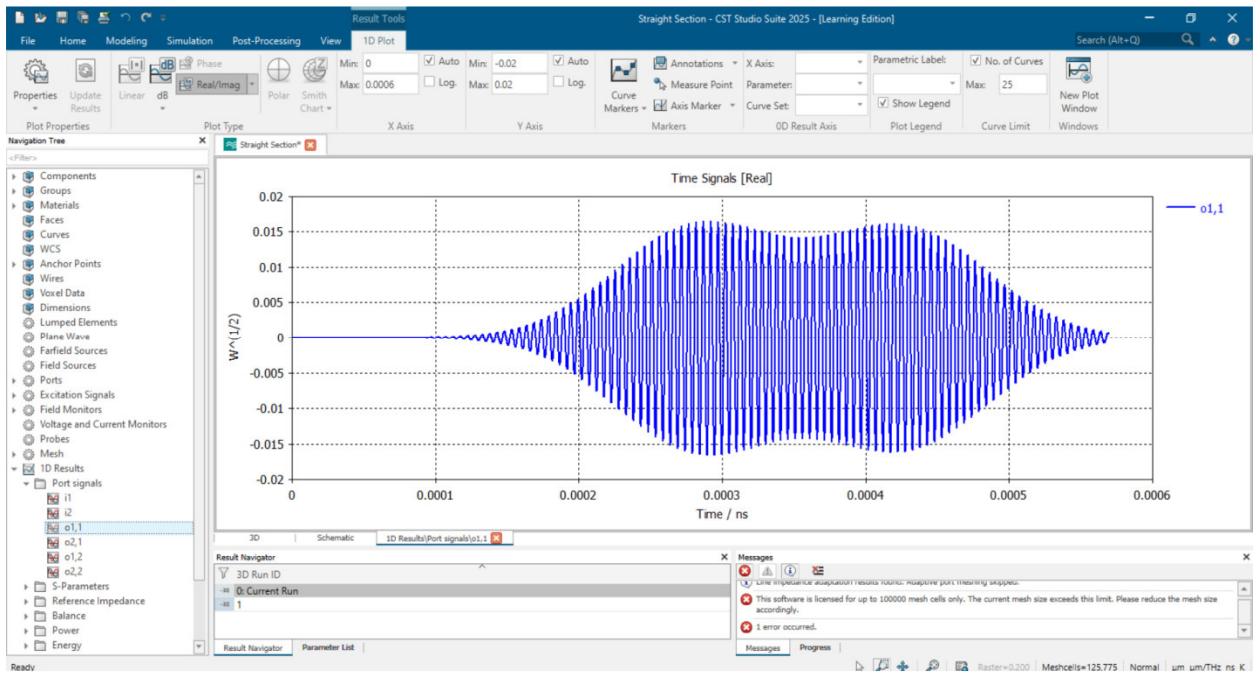
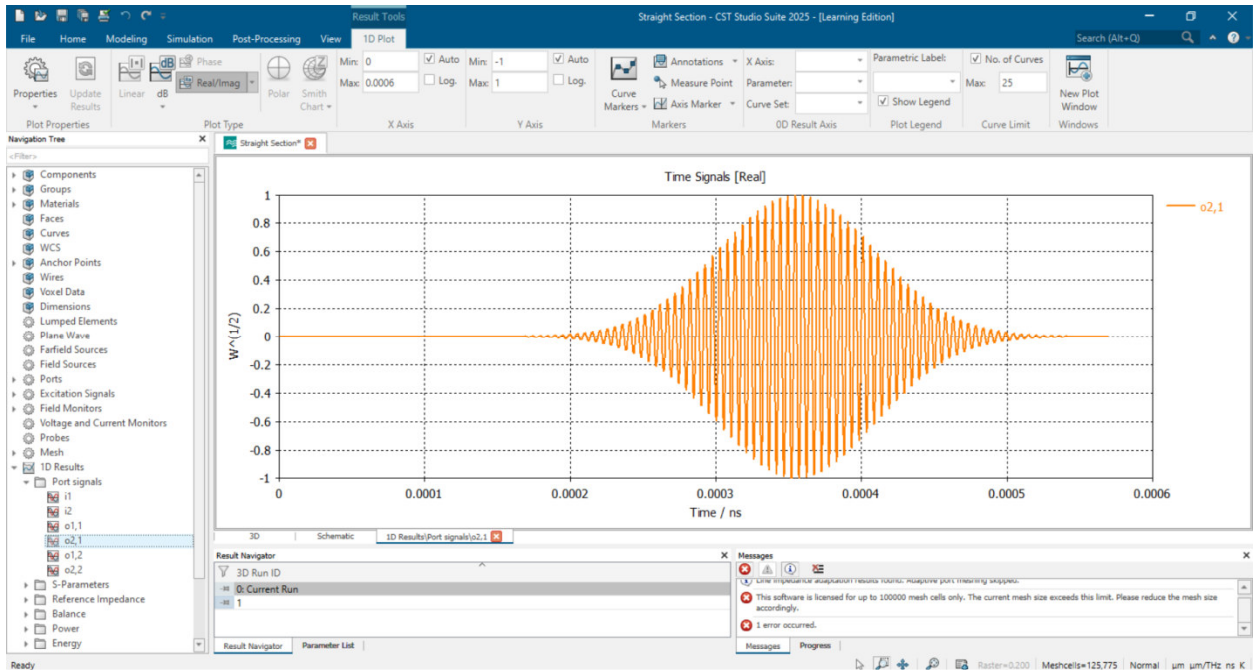
$$S_{12} = \left. \frac{b_1}{a_2} \right|_{a_1=0}$$

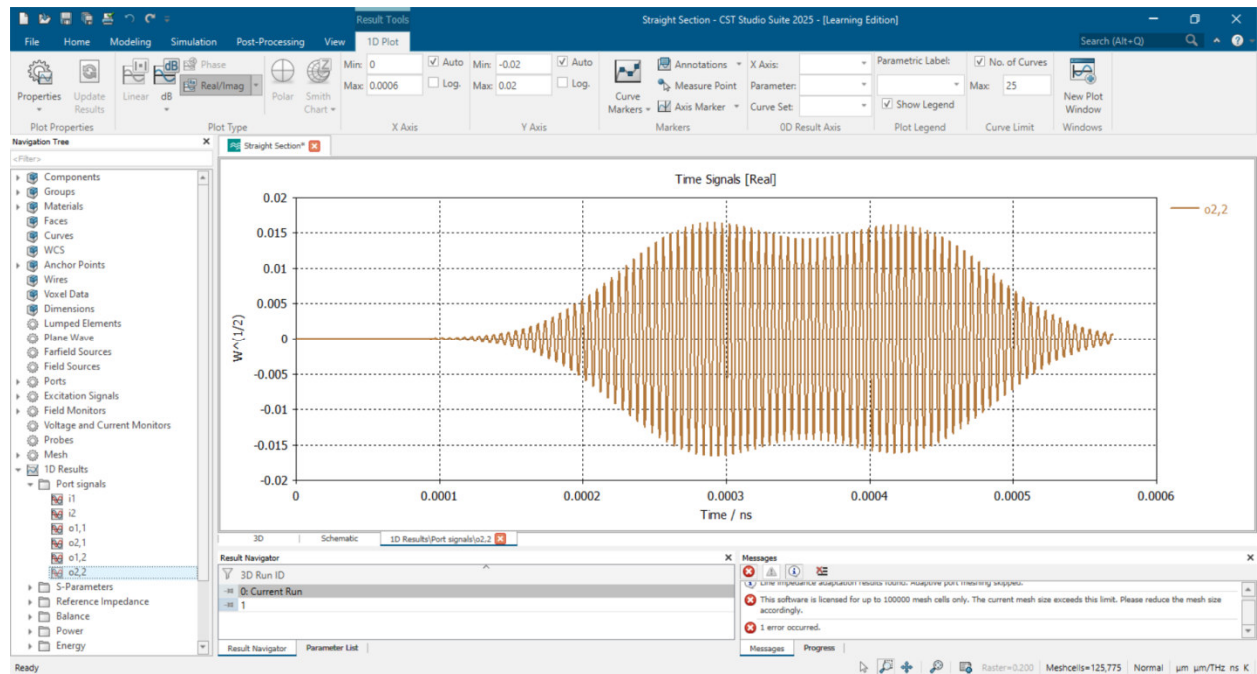
Output Reflection Coefficient –  $S_{22}$

$$S_{22} = \left. \frac{b_2}{a_2} \right|_{a_1=0}$$

**OUTPUT:**







| Port (i-input,<br>o-output) | Time (ns) | Peak Power $W^{1/2}$ |
|-----------------------------|-----------|----------------------|
| <b>o1,2</b>                 |           |                      |
| <b>o2,1</b>                 |           |                      |
| <b>o1,1</b>                 |           |                      |
| <b>o2,2</b>                 |           |                      |

## RESULT:

The rectangular waveguide model was successfully simulated using CST Studio Suite over the specified microwave frequency range. Simulation of the power distribution through the ports validates theoretical waveguide behavior, field distribution characteristics, cutoff frequency, and transmission performance of the rectangular waveguide.

## Test Case 2:

To simulate a Rectangular Waveguide using CST Studio Suite, analyze its electromagnetic behavior, and examine the S parameters.

**AIM:** To simulate a Rectangular Waveguide using CST Studio Suite, analyze its electromagnetic behavior, and examine S parameters.

## TOOL REQUIRED:

CST Studio Suite

## PROCEDURE:

Open CST Studio Suite

Open file-new

Create a rectangular waveguide section

Define ports

Run the simulation.

Observe:

- $S_{11}$  (Reflection Coefficient)
- $S_{21}$  (Transmission Coefficient)

Plot: S-parameter vs Frequency

## CALCULATION

$$P_{reflected} = |S_{11}|^2 P_{input}$$

$$P_{transmitted} = |S_{21}|^2 P_{input}$$

$$P_{loss} = P_{input} - (P_{reflected} + P_{transmitted})$$

Two-Port Network S-Parameter Matrix

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

Input Reflection Coefficient –  $S_{11}$

$$S_{11} = \left. \frac{b_1}{a_1} \right|_{a_2=0}$$

Forward Transmission Coefficient –  $S_{21}$

$$S_{21} = \left. \frac{b_2}{a_1} \right|_{a_2=0}$$

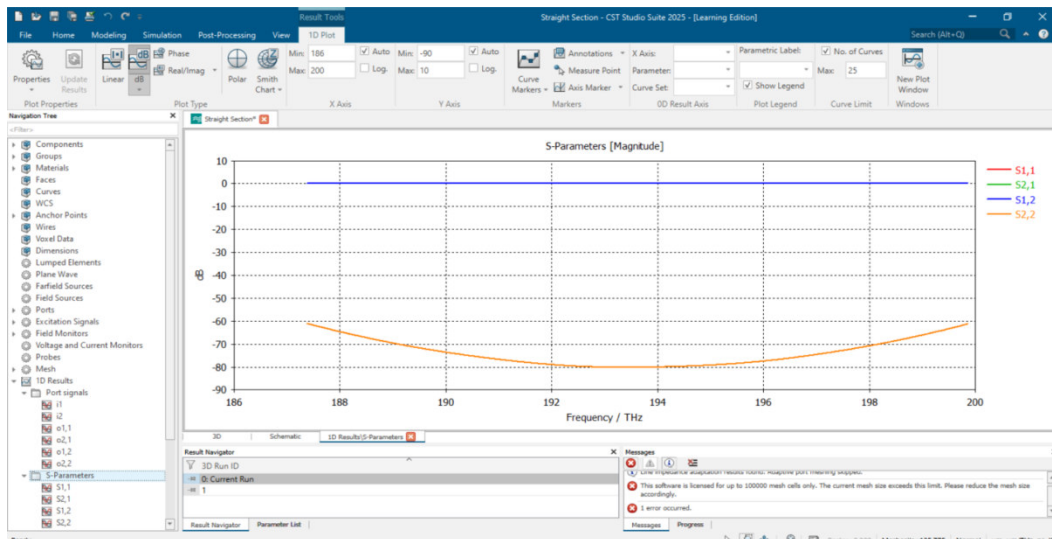
Reverse Transmission Coefficient –  $S_{12}$

$$S_{12} = \left. \frac{b_1}{a_2} \right|_{a_1=0}$$

Output Reflection Coefficient –  $S_{22}$

$$S_{22} = \left. \frac{b_2}{a_2} \right|_{a_1=0}$$

**OUTPUT:**



|            | Frequency (GHz) | S Parameters<br>Magnitude (dB) |
|------------|-----------------|--------------------------------|
| <b>S11</b> |                 |                                |
| <b>S21</b> |                 |                                |
| <b>S12</b> |                 |                                |
| <b>S22</b> |                 |                                |

**RESULT:**

The rectangular waveguide model was successfully simulated using CST Studio Suite over the specified microwave frequency range. Simulation of S parameters validates theoretical waveguide behavior, field distribution characteristics, cutoff frequency, and transmission performance of the rectangular waveguide.

|                 |                                                                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EXNO: 12</b> | <b>Performance Evaluation of a Linear Tapered Rectangular Waveguide in terms of power distribution analysis per ports and S-parameters.</b> |
| <b>DATE:</b>    |                                                                                                                                             |

### **Test Case 1:**

Simulate a linear tapered Rectangular Waveguide using CST Studio Suite, analyze its electromagnetic behavior, and examine the power distribution among ports over a specified frequency range.

**AIM:** To simulate a linear tapered Rectangular Waveguide using CST Studio Suite, analyze its electromagnetic behavior, and examine the power distribution among ports over a specified frequency range.

### **TOOL REQUIRED:**

CST Studio Suite

### **PROCEDURE:**

Open CST Studio Suite  
 Open file-new  
 Create a linear tapered rectangular waveguide  
 Define ports  
 Run the simulation.  
 Observe:

- S11 (Reflection Coefficient)
- S21 (Transmission Coefficient)
- Power accepted at Port 1
- Power transmitted to Port 2

Plot the Power distribution at ports

### **CALCULATION**

$$P_{reflected} = |S_{11}|^2 P_{input}$$

$$P_{transmitted} = |S_{21}|^2 P_{input}$$

$$P_{loss} = P_{input} - (P_{reflected} + P_{transmitted})$$

## Two-Port Network S-Parameter Matrix

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

Input Reflection Coefficient –  $S_{11}$

$$S_{11} = \left. \frac{b_1}{a_1} \right|_{a_2=0}$$

Forward Transmission Coefficient –  $S_{21}$

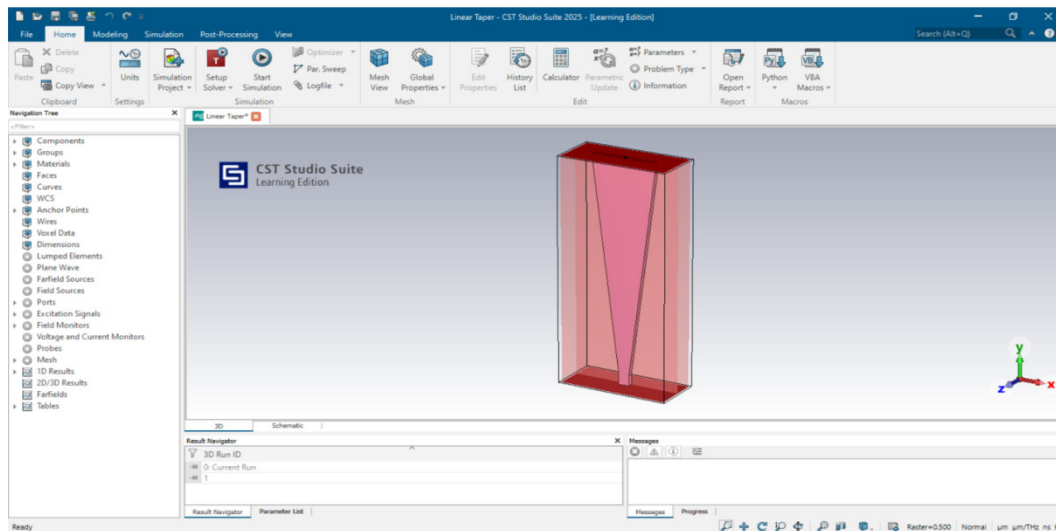
$$S_{21} = \left. \frac{b_2}{a_1} \right|_{a_2=0}$$

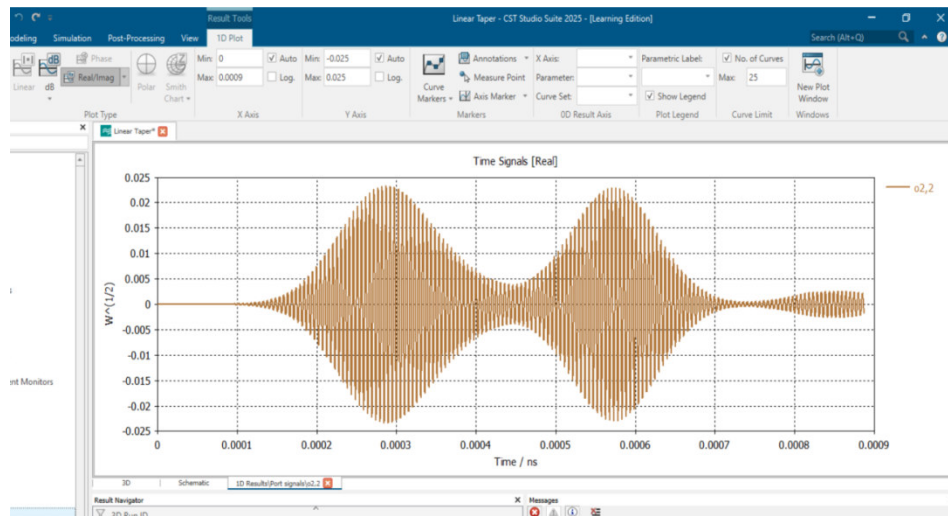
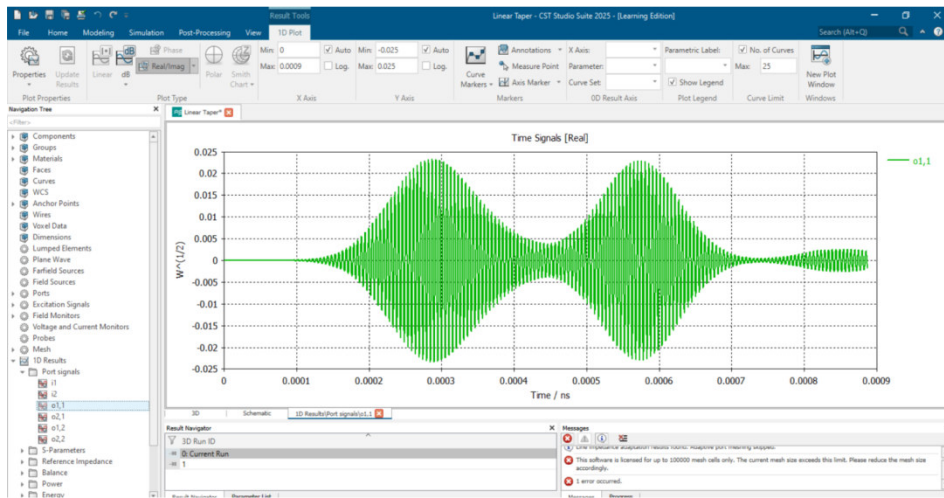
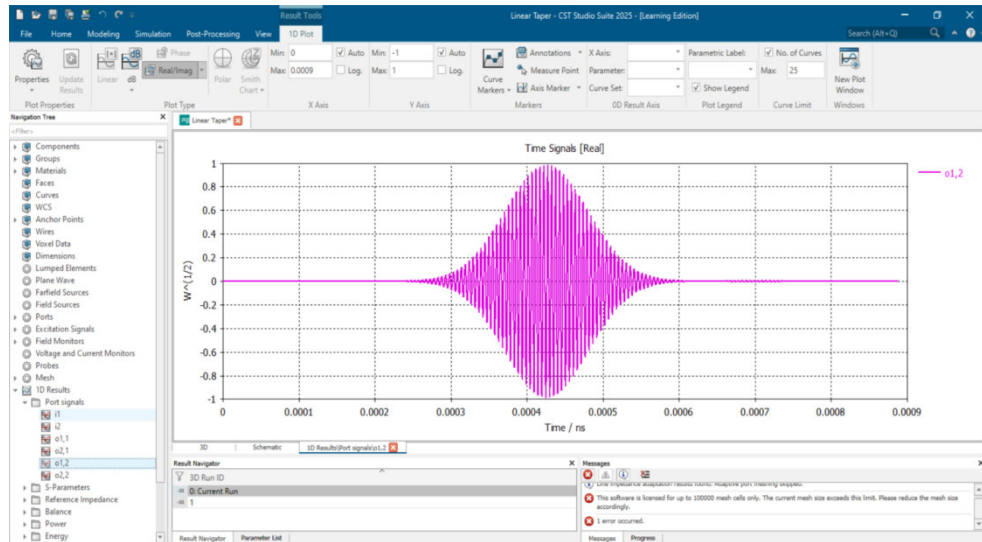
Reverse Transmission Coefficient –  $S_{12}$

$$S_{12} = \left. \frac{b_1}{a_2} \right|_{a_1=0}$$

Output Reflection Coefficient –  $S_{22}$

$$S_{22} = \left. \frac{b_2}{a_2} \right|_{a_1=0}$$





| Port (i-input,<br>o-output) | Time (ns) | Peak Power $W^{1/2}$ |
|-----------------------------|-----------|----------------------|
| <b>o1,2</b>                 |           |                      |
| <b>o2,1</b>                 |           |                      |
| <b>o1,1</b>                 |           |                      |
| <b>o2,2</b>                 |           |                      |

## RESULT:

The power distribution of the linear tapered rectangular waveguide was analyzed over the specified frequency range. The transmission performance confirms efficient signal propagation through the waveguide, demonstrating stable electromagnetic performance across the operating band.

## Test Case 2:

Simulate a linear tapered Rectangular Waveguide using CST Studio Suite, analyze its electromagnetic behavior, and examine the S parameters.

**AIM:** To simulate a linear tapered Rectangular Waveguide using CST Studio Suite, analyze its electromagnetic behavior, and examine the S parameters.

## TOOL REQUIRED:

CST Studio Suite

## PROCEDURE:

Open CST Studio Suite

Open file-new

Create a linear tapered rectangular waveguide

Define ports

Run the simulation.

Observe:

- S11 (Reflection Coefficient)
- S21 (Transmission Coefficient)

Plot: S-parameter vs Frequency

## CALCULATION

$$P_{reflected} = |S_{11}|^2 P_{input}$$

$$P_{transmitted} = |S_{21}|^2 P_{input}$$

$$P_{loss} = P_{input} - (P_{reflected} + P_{transmitted})$$

Two-Port Network S-Parameter Matrix

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

Input Reflection Coefficient –  $S_{11}$

$$S_{11} = \left. \frac{b_1}{a_1} \right|_{a_2=0}$$

Forward Transmission Coefficient –  $S_{21}$

$$S_{21} = \left. \frac{b_2}{a_1} \right|_{a_2=0}$$

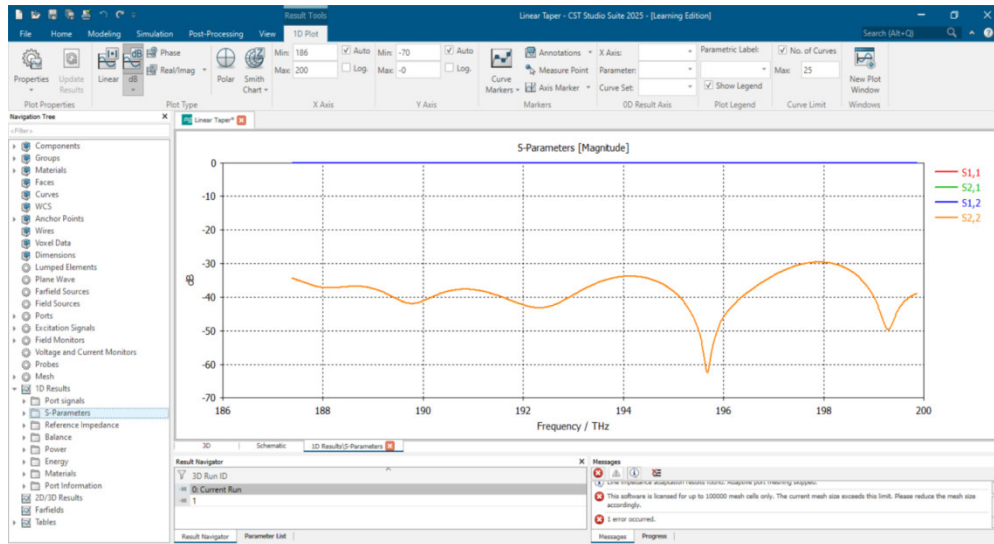
Reverse Transmission Coefficient –  $S_{12}$

$$S_{12} = \left. \frac{b_1}{a_2} \right|_{a_1=0}$$

Output Reflection Coefficient –  $S_{22}$

$$S_{22} = \left. \frac{b_2}{a_2} \right|_{a_1=0}$$

**OUTPUT:**



|            | Frequency (GHz) | S Parameters<br>Magnitude (dB) |
|------------|-----------------|--------------------------------|
| <b>S11</b> |                 |                                |
| <b>S21</b> |                 |                                |
| <b>S12</b> |                 |                                |
| <b>S22</b> |                 |                                |

## RESULT:

The S-parameters of the linear tapered rectangular waveguide were analyzed over the specified frequency range. The transmission performance confirms efficient signal propagation through the waveguide, demonstrating stable electromagnetic performance across the operating band.

|                 |                                                                                                             |
|-----------------|-------------------------------------------------------------------------------------------------------------|
| <b>EXNO: 13</b> | <b>Investigation of Electromagnetic Field distribution in Rectangular to Circular Waveguide transition.</b> |
| <b>DATE:</b>    |                                                                                                             |

### Test Case 1:

Analyze the electric field and magnetic field distribution at Port 1 of the rectangular waveguide to circular waveguide transition and examine the electric and magnetic field parameters at Port 1 for studying electromagnetic field behavior and validating the system performance.

**AIM:** To analyze the electric field and magnetic field distribution at Port 1 of the circular resonator and examine the electric and magnetic field parameters at Port 1 for studying electromagnetic field behavior and validating the system performance.

### TOOL REQUIRED:

CST Studio Suite

### PROCEDURE:

Open CST Studio Suite  
 Open file-new  
 Create a rectangular to circular waveguide transition  
 Define ports  
 Run the simulation.  
 Observe:

- Electric field distribution at port 1 (rectangular waveguide)
- Magnetic field distribution at port 1 (rectangular waveguide)
- Measure the Electric and magnetic field parameters at port 1.

### CALCULATION:

#### 1. Rectangular Waveguide

For a rectangular waveguide with dimensions

- $a$  = width
- $b$  = height

The cutoff frequency for the  $\text{TE}_{mn}$  or  $\text{TM}_{mn}$  mode is:

$$f_c = \frac{1}{2\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

For air-filled waveguide ( $\mu = \mu_0, \epsilon = \epsilon_0$ ):

$$f_c = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

## 1. Rectangular Waveguide

For a rectangular waveguide with dimensions

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For air-filled waveguide ( $\mu = \mu_0, \varepsilon = \varepsilon_0$ ):

$$f_c = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

## 2. Circular Waveguide

For a circular waveguide with radius  $a$ :

The cutoff frequency for  $\text{TE}_{mn}$  mode:

$$f_c = \frac{X'_{mn}}{2\pi a \sqrt{\mu\varepsilon}}$$

The cutoff frequency for  $\text{TM}_{mn}$  mode:

$$f_c = \frac{X_{mn}}{2\pi a \sqrt{\mu\varepsilon}}$$

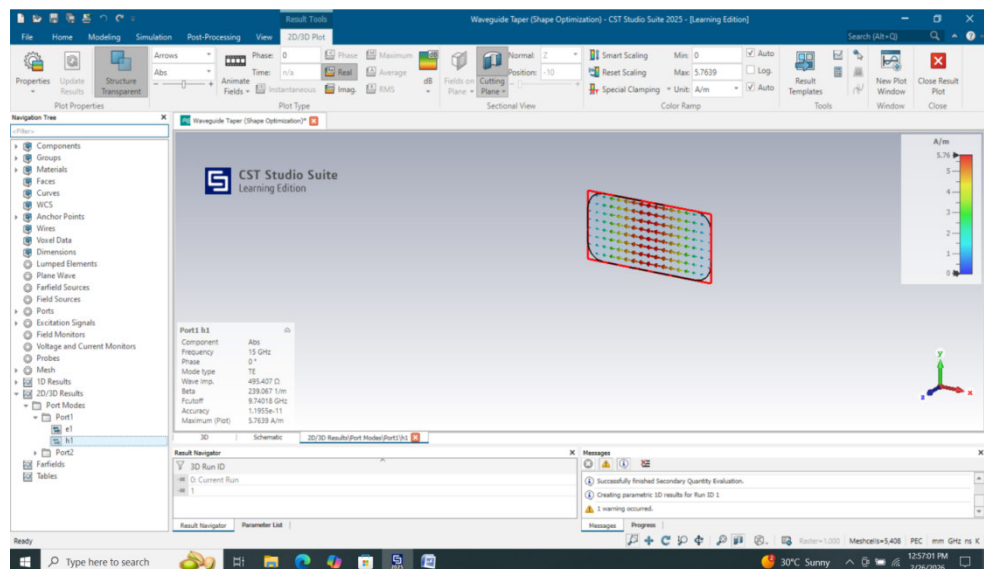
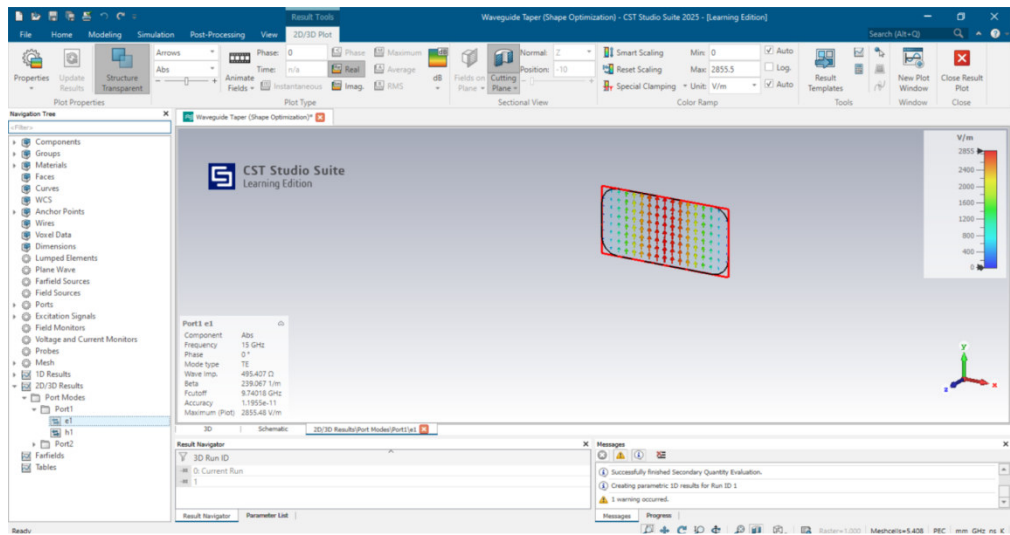
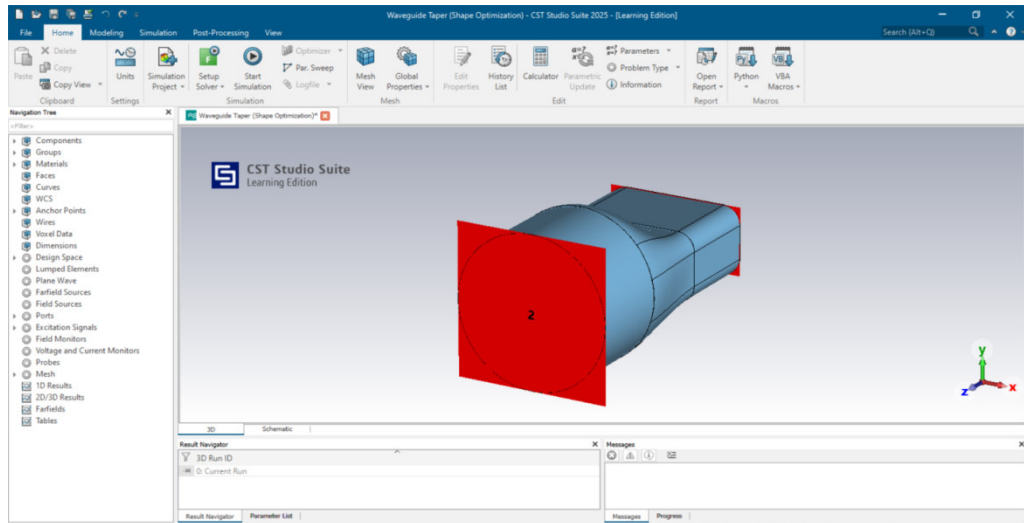
For air-filled waveguide:

$$f_c = \frac{X'_{mn}c}{2\pi a} \quad (\text{TE modes})$$
$$f_c = \frac{X_{mn}c}{2\pi a} \quad (\text{TM modes})$$

Where:

- $X'_{mn} = n^{\text{th}}$  root of derivative of Bessel function (TE modes)
- $X_{mn} = n^{\text{th}}$  root of Bessel function (TM modes)
- $c$  = speed of light
- $\mu$  = permeability
- $\varepsilon$  = permittivity

**OUTPUT:**



| Parameter              | Port 1              |                     |
|------------------------|---------------------|---------------------|
|                        | e1 (electric field) | h1 (magnetic field) |
| Frequency              |                     |                     |
| Mode type              |                     |                     |
| Wave impedance         |                     |                     |
| Phase constant $\beta$ |                     |                     |
| Cutoff frequency       |                     |                     |
| Peak Value             |                     |                     |

## RESULT:

The rectangular to circular waveguide transition was successfully simulated using CST Studio Suite. The electric field distribution at Port 1 was observed and analyzed, showing the expected spatial variation and field intensity profile across the port region. The magnetic field distribution at Port 1 was also obtained, indicating the corresponding magnetic field patterns consistent with electromagnetic wave propagation. The electric and magnetic field parameters at Port 1 were successfully measured and recorded.

## Test Case 2:

Analyze the electric field and magnetic field distribution at Port 2 of the rectangular waveguide to circular waveguide transition and examine the electric and magnetic field parameters at Port 2 for studying electromagnetic field behavior and validating the system performance.

**AIM:** To analyze the electric field and magnetic field distribution at Port 2 of the circular resonator and examine the electric and magnetic field parameters at Port 2 for studying electromagnetic field behavior and validating the system performance.

## TOOL REQUIRED:

CST Studio Suite

## PROCEDURE:

Open CST Studio Suite

Open file-new

Create a rectangular to circular waveguide transition

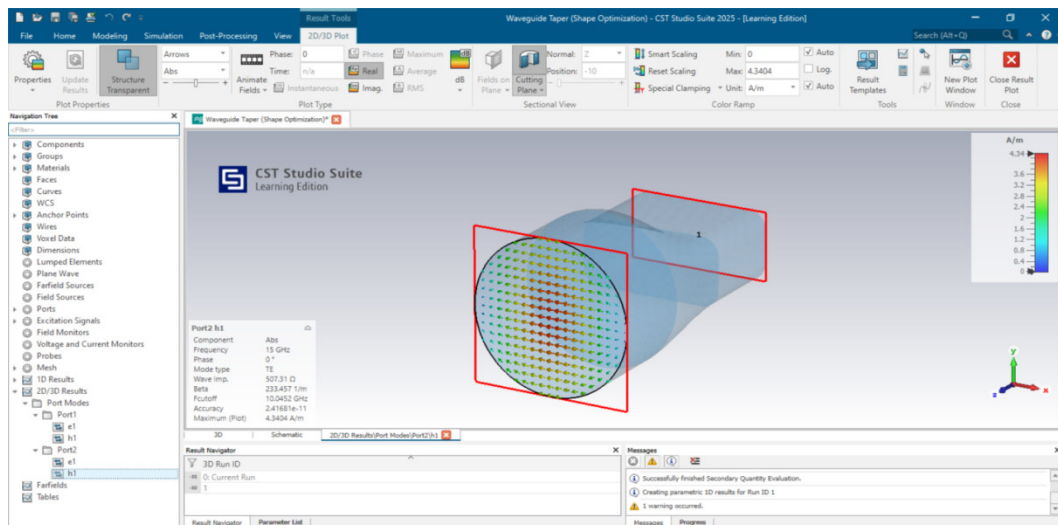
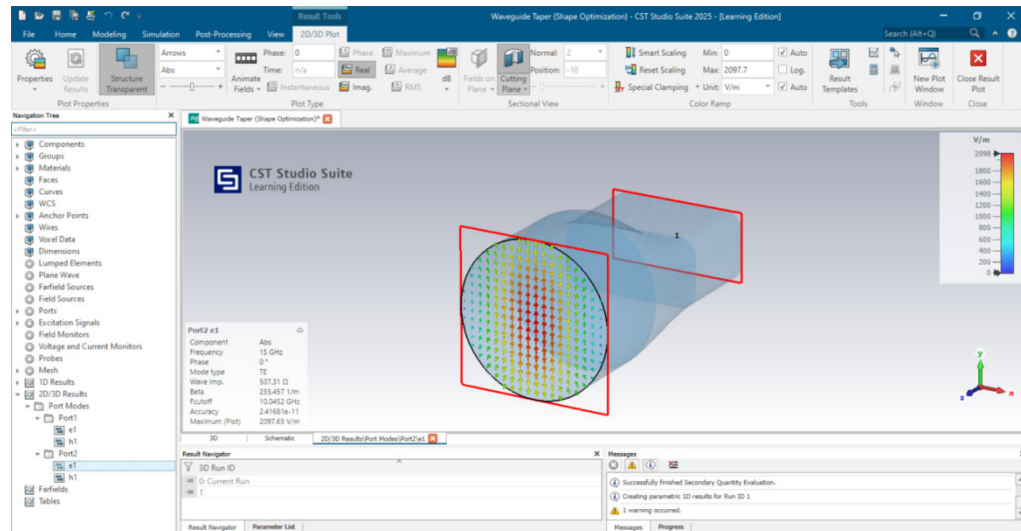
Define ports

Run the simulation.

Observe:

- Electric field distribution at port 2 (circular waveguide)
- Magnetic field distribution at port 2 (circular waveguide)
- Measure the Electric and magnetic field parameters at port 2.

**OUTPUT:**



**OUTPUT:**

| Parameter              | Port 2              |                     |
|------------------------|---------------------|---------------------|
|                        | e2 (electric field) | h2 (magnetic field) |
| Frequency              |                     |                     |
| Mode type              |                     |                     |
| Wave impedance         |                     |                     |
| Phase constant $\beta$ |                     |                     |
| Cutoff frequency       |                     |                     |
| Peak Value             |                     |                     |

**RESULT:**

The rectangular to circular waveguide transition was successfully simulated using CST Studio Suite. The electric field distribution at Port 2 was observed and analyzed, showing the expected spatial variation and field intensity profile across the port region. The magnetic field distribution at Port 2 was also obtained, indicating the corresponding magnetic field patterns consistent with electromagnetic wave propagation. The electric and magnetic field parameters at Port 2 were successfully measured and recorded.

|                 |                                                                                                           |
|-----------------|-----------------------------------------------------------------------------------------------------------|
| <b>EXNO: 14</b> | <b>Electromagnetic Field and Power Distribution Analysis of a Circular Resonator in CST Studio Suite.</b> |
| <b>DATE:</b>    |                                                                                                           |

### **Test Case 1:**

Analyze the electric field and magnetic field distribution at Port 1 of the circular resonator and examine the electric and magnetic field parameters at Port 1 for studying electromagnetic field behavior and validating the system performance.

**AIM:** To analyze the electric field and magnetic field distribution at Port 1 of the circular resonator and examine the electric and magnetic field parameters at Port 1 for studying electromagnetic field behavior and validating the system performance.

### **TOOL REQUIRED:**

CST Studio Suite

### **PROCEDURE:**

Open CST Studio Suite

Open file-new

Design a circular resonator

Define ports

Run the simulation.

Observe:

- Electric field distribution at port 1
- Magnetic field distribution at port 1
- Measure the Electric and magnetic field parameters at port 1 .

## CALCULATION:

### Line Impedance (Characteristic Impedance) – Circular Resonator

For a circular waveguide resonator (radius  $a$ ), the **line impedance** (or characteristic impedance) depends on the mode of propagation.

For  $TE_{mn}$  mode:

$$Z_{TE} = \frac{\omega\mu}{\beta}$$

For  $TM_{mn}$  mode:

$$Z_{TM} = \frac{\beta}{\omega\varepsilon}$$

Where:

- $\omega = 2\pi f$
- $\mu$  = permeability
- $\varepsilon$  = permittivity
- $\beta = \sqrt{k^2 - k_c^2}$  (phase constant)
- $k = \omega\sqrt{\mu\varepsilon}$
- $k_c$  = cutoff wave number

### Wave Impedance – Circular Resonator

Wave impedance inside a circular resonator (or circular waveguide cavity) is:

For **TE mode**:

$$\eta_{TE} = \frac{\eta}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$$

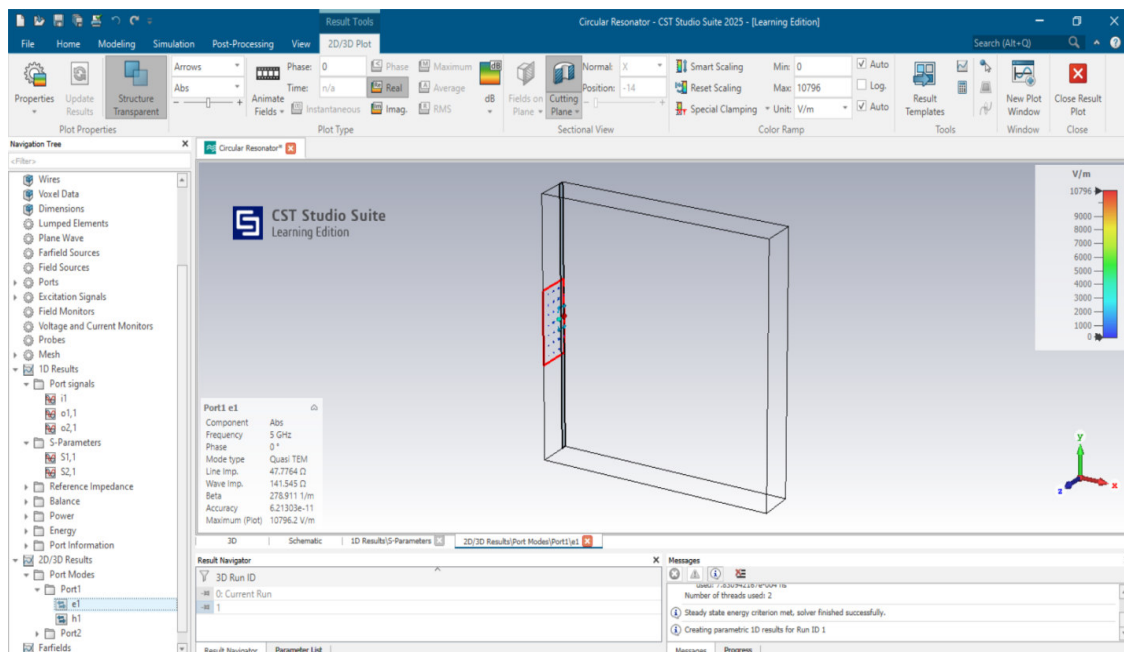
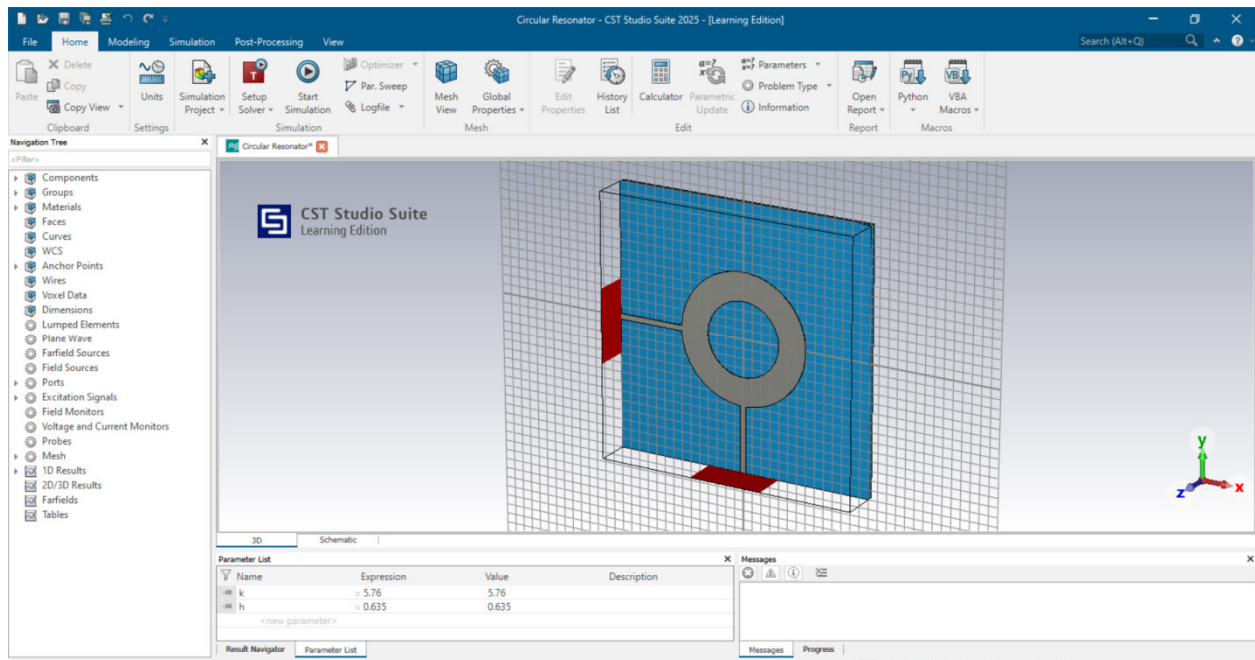
For **TM mode**:

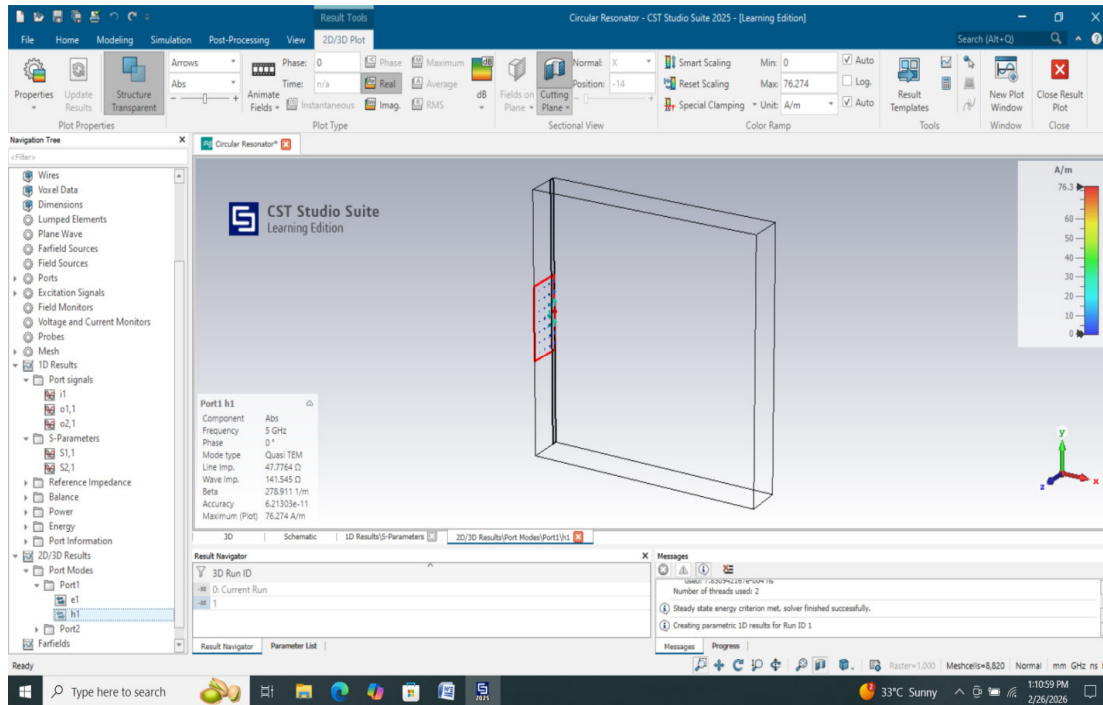
$$\eta_{TM} = \eta \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

Where:

- $\eta = \sqrt{\frac{\mu}{\varepsilon}}$  (intrinsic impedance of medium)
- $f_c$  = cutoff frequency
- $f$  = operating frequency

## OUTPUT:





| Parameter              | Port 1              |                     |
|------------------------|---------------------|---------------------|
|                        | e2 (electric field) | h2 (magnetic field) |
| Frequency              |                     |                     |
| Mode type              |                     |                     |
| Wave impedance         |                     |                     |
| Line impedance         |                     |                     |
| Phase constant $\beta$ |                     |                     |
| Peak Value             |                     |                     |

## **RESULT:**

The circular resonator was successfully simulated using CST Studio Suite. The electric field distribution at Port 1 was observed and analyzed, showing the expected spatial variation and field intensity profile across the port region. The magnetic field distribution at Port 1 was also obtained, indicating the corresponding magnetic field patterns consistent with electromagnetic wave propagation. The electric and magnetic field parameters at Port 1 were successfully measured and recorded.

## **Test Case 2:**

Analyze the electric field and magnetic field distribution at Port 2 of the circular resonator and examine the electric and magnetic field parameters at Port 2 for studying electromagnetic field behavior and validating the system performance.

**AIM:** To analyze the electric field and magnetic field distribution at Port 2 of the circular resonator and examine the electric and magnetic field parameters at Port 2 for studying electromagnetic field behavior and validating the system performance.

## **TOOL REQUIRED:**

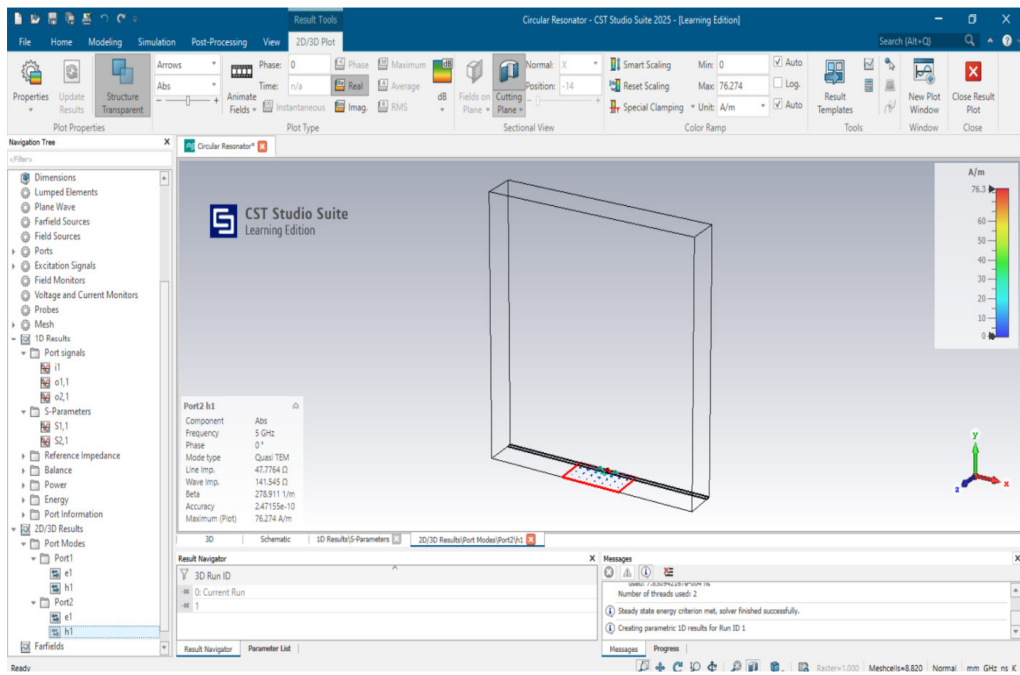
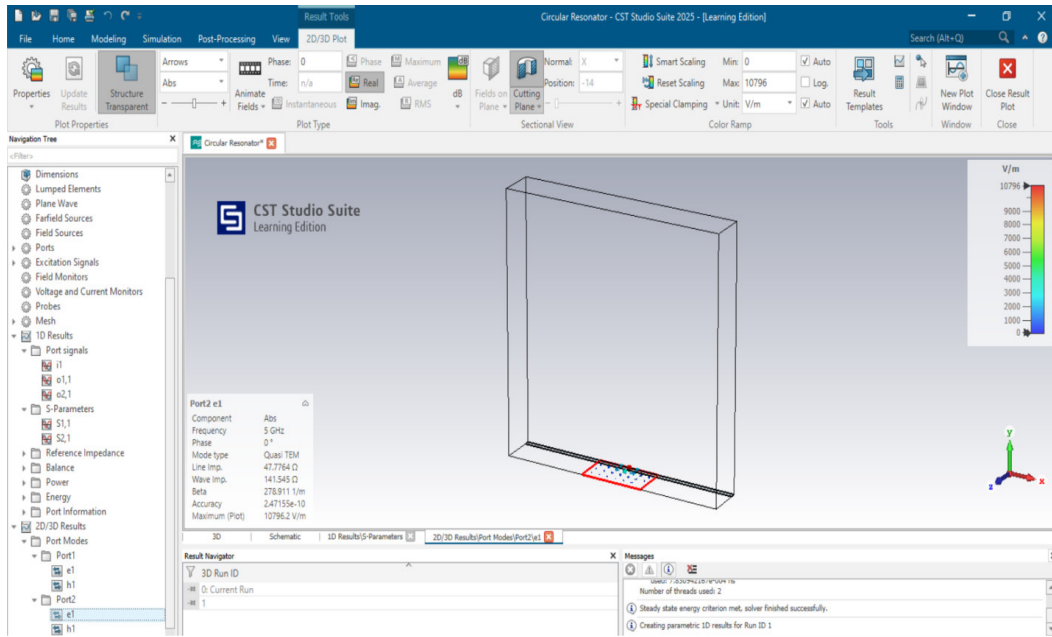
CST Studio Suite

## **PROCEDURE:**

Open CST Studio Suite  
Open file-new  
Design a circular resonator  
Define ports  
Run the simulation.  
Observe:

- Electric field distribution at port 2
- Magnetic field distribution at port 2
- Measure the Electric and magnetic field parameters at port 2.

## **OUTPUT:**



| Parameter | Port 2              |                     |
|-----------|---------------------|---------------------|
|           | e2 (electric field) | h2 (magnetic field) |
| Frequency |                     |                     |
| Mode type |                     |                     |

|                                          |  |  |
|------------------------------------------|--|--|
| <b>Wave impedance</b>                    |  |  |
| <b>Line impedance</b>                    |  |  |
| <b>Phase constant <math>\beta</math></b> |  |  |
| <b>Peak Value</b>                        |  |  |

## RESULT:

The circular resonator was successfully simulated using CST Studio Suite. The electric field distribution at Port 2 was observed and analyzed, showing the expected spatial variation and field intensity profile across the port region. The magnetic field distribution at Port 2 was also obtained, indicating the corresponding magnetic field patterns consistent with electromagnetic wave propagation. The electric and magnetic field parameters at Port 2 were successfully measured and recorded.